Religion, Prayer and Health Production

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Abstract

This paper tests whether a religious practice behaves as an input into health production. In the US, most adults diagnosed with a range of common illnesses believe that prayer is an effective treatment and have prayed for recovery. We study the COVID-19 pandemic, when most people believed prayer was protective. We embed prayer in a health production model with vaccination as a substitute input. We then test the model's predictions using staggered vaccine eligibility as a source of plausibly exogenous variation in the price of vaccination. Vaccine eligibility reduces weekly prayer just as it reduces other protective behaviors. As predicted, the decline is concentrated among weakly religious people, for whom prayer is marginal, and is near zero among strongly religious people, for whom it is inframarginal. These patterns align with perceived health production and not simply with a coping response to distress. The vaccination response does not vary by religiosity.

JEL: I12, Z12, I18

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1 Introduction

Economic models typically assume that people optimize using objective production technologies. However, in many domains, such as health, finance, and education, behavior depends upon subjective beliefs about how actions translate into outcomes. Cultural and religious institutions often shape these beliefs. An unanswered question is whether behaviors rooted in these belief systems respond to relative prices in the way that standard inputs do.

Prayer is a prominent example of such a behavior. Many people ask God or another higher power for help with life challenges and believe that these petitions are effective (Upenieks 2023). Health is a common topic of prayer (Wilkinson et al. 2008). In Figure 1, majorities of US adults diagnosed with a range of common illnesses believe that prayer is an effective treatment and report having prayed for recovery.[1] Research has interpreted increases in prayer during adverse events as a coping response that produces comfort (Bentzen 2019, Bentzen 2021). If prayer also functions as a health input, reliance on prayer should respond to the price of substitutes.

We study this question in the context of COVID-19 in the United States. As with other illnesses, majorities of US adults saw prayer as effective against COVID-19 and prayed regularly to stay safe. We develop a simple model in which people produce perceived health using vaccines and another input, such as prayer. We then derive how the arrival of vaccines should affect prayer, avoidance (e.g., social distancing and the use of masks), and perceived health risk, and how these effects should vary with religiosity and with beliefs about vaccine effectiveness. We test these predictions using longitudinal data from the Understanding America Study, a nationally representative panel survey conducted throughout the COVID-19 pandemic (Alattar et al. 2018, Kapteyn et al. 2020). From December 2020 through April 2021, the government rolled out vaccines that greatly reduced the price of this input from infinity to near zero. We exploit variation in vaccine eligibility across individuals and over time to estimate the impact of eligibility on vaccination, perceived mortality risk, prayer, and alternative protective behaviors.

[1] Across nine of ten listed illnesses, a majority believe prayer is effective and have prayed. The figure is based on a survey from December 2024 to February 2025 of 1814 participants in the Understanding America Study. Participants were drawn from a representative sample and had received the listed diagnoses within the past twelve months.

Vaccine eligibility sharply increases vaccination and reduces perceived mortality risk. Consistent with substitution across perceived health inputs, it also reduces the probability of weekly prayer by 0.9 percentage points (p.p.; 1.5 percent) and lowers avoidance by 0.05 standard deviations (s.d.). Scaling by the impact on vaccination yields Wald estimates of -6.8 p.p. for prayer and -0.38 s.d. for avoidance among those induced to vaccinate. All four point estimates are larger for people who believe that the vaccine is effective; they are small for people who doubt the vaccine, which suggests that the response operates through beliefs about the vaccine rather than through other factors that may coincide with the vaccine rollout.

We next examine heterogeneity by religiosity. If prayer functions as a perceived health input, then its responsiveness to vaccine eligibility should depend on its centrality to a person's religious practice. The decline in prayer occurs primarily among weakly religious people, for whom vaccine eligibility reduces prayer by 1.6 p.p. (26 percent). The effect is small and statistically insignificant for strongly religious people. Distinguishing between types of weakly religious people further clarifies the mechanism. Atheists and agnostics, who are 26 percent of the weakly religious subgroup, pray infrequently both before and after eligibility. The decline in prayer is concentrated among *other* weakly religious people. Prayer is plausibly a marginal input for these people, many of whom are sympathetic to religion despite their weak attachment. By contrast, the impact on vaccination is similar for strongly and weakly religious respondents (14 and 15 p.p.).

This pattern of heterogeneity also distinguishes substitution from coping, under which people pray to relieve distress. The literature suggests that religious coping is concentrated among strongly religious people (Pargament 2001, Bentzen 2021), whereas we observe an effect on prayer primarily among weakly religious people. In addition, we do not find that vaccine eligibility reduces anxiety or improves perceived control, as it would if prayer were falling because distress had eased. These patterns favor the substitution interpretation.

This paper makes three contributions. First, it extends the economics of religion by modeling a religious practice as an input into a production technology and testing the comparative statics of that model. Prior work models religious participation as an optimizing choice (Azzi and Ehrenberg 1975, Iannaccone 1990) and shows that religious activity re-

sponds to the cost of participation (Gruber and Hungerman 2008), to the availability of insurance (Chen 2010, Auriol et al. 2020), and to the returns to secular relative to religious activity (Cantoni et al. 2018, Becker et al. 2016). We treat prayer as one of several inputs that produce perceived health, so that its use depends on its perceived effectiveness and relative prices. The size of the decline in prayer depends on the weight people place on prayer relative to the substitute, a relationship that prior studies do not test. As the model predicts, the decline is larger among people who believe the vaccine is effective and among people who are weakly religious.

Second, the heterogeneity in our estimates shows that the same practice can be marginal for some people and inframarginal for others. Weakly religious people who are not atheists or agnostics adjust prayer when a substitute becomes cheaper. The strongly religious, who see prayer as relatively productive, do not appear to respond. Actions that are central to identity may be inframarginal in this way, while more peripheral practices respond to changes in relative prices.

Finally, the vaccine rollout provides a setting to examine substitution across protective behaviors in response to the arrival of an alternative protective technology. We find evidence of substitution between vaccination and prayer and between vaccination and avoidance. The vaccination response is similar for weakly and strongly religious people. These findings contribute to debates on behavioral responses to safety innovations (Peltzman 1975, Kaestner et al. 2014, Buckell et al. 2023) and to evidence about religiosity and technology adoption (Squicciarini 2020).

2 Context and Data

2.1 Pandemic Timing and Policies

In 2020, in response to the COVID-19 pandemic, the US implemented Operation Warp Speed to accelerate the development of COVID-19 vaccines (Burgos et al. 2021). The Food and Drug Administration granted an emergency use authorization (EUA) for the Pfizer vaccine on December 11, 2020. The leading vaccines were most effective at reducing the complications associated with severe disease but also helped to prevent infection (Mohammed et al. 2022).

Contemporaneous news accounts show that the public quickly became aware of this feature of the vaccines (Medical News Today 2021).

Given the limited initial supply of vaccines, health authorities implemented a phased vaccination campaign that led to plausibly exogenous changes in access to vaccines. This effort loosely followed guidelines developed by the CDC Advisory Committee on Immunization Practices (ACIP) to offer the vaccine first to vulnerable people and essential workers (Dooling 2021). In general, the guidelines prioritized first responders (defined in various ways), the elderly, and people with existing health vulnerabilities. However, states set their own eligibility schedules, adapting the guidelines to local priorities and circumstances. Eligibility dates also varied according to vaccine availability within each jurisdiction. Vaccinations began in earnest in January of 2021 and proceeded throughout the winter and spring of that year. By mid-April 2021, everyone aged 16 or older was eligible for vaccination. Although eligibility rules varied by state, age and health status explain most of the variation in when people became eligible.

Figure 2 illustrates the timing of the vaccine rollout and the importance of vaccine eligibility. We divide the sample by the median eligibility date and plot the percent of people who have been vaccinated by date for those with early and late eligibility. Red vertical bars in the figure indicate the beginning and end of the vaccine rollout. Those who were eligible early received the vaccine sooner. By February 1, 2021, those with early eligibility were 11 p.p. more likely to be vaccinated than those with late eligibility. By March 1, this difference rose to 23 p.p. and was 18 p.p. by April 1.[2]

2.2 Data and Variables

This analysis relies on data from the Understanding America Study (UAS). The UAS is an online panel survey administered by the University of Southern California (Alattar et al. 2018). Panel members are selected through address-based sampling and are compensated for participation. The survey uses post-stratification weights to make the sample nationally

[2]The difference persists into the summer of 2021, after all adults had become eligible. This pattern likely reflects higher take-up among older adults, who became eligible earlier. It does not bear on identification, which relies on comparisons made while some respondents are not yet eligible and requires that people with early and late eligibility would have followed parallel trends in the absence of eligibility. Section 4 assesses this assumption.

representative in terms of sex, race, age, marital status, and education. Weights also adjust for differences in the sampling probabilities of geographic units according to Census region, urbanicity, and population size. The number of respondents has expanded over time, to around 9100 active participants in 2023.

During the pandemic, the UAS implemented a high-frequency panel survey related to COVID-19. Surveyors followed up with respondents 34 times from March 2020 through July 2023, with most surveys occurring every two weeks during 2020 and early 2021 (Kapteyn et al. 2020).[3] The COVID-19 panel survey included an extensive array of consistently measured questions related to the health and economic impact of the pandemic.

To construct vaccine eligibility, our primary independent variable, we combine publicly available vaccine eligibility rules with individual respondent characteristics. Individual-specific eligibility dates are based on state of residence, age, occupation, chronic health conditions, and nursing home residence.[4] Because of variation in the details of the rollout by state, we identify 64 distinct eligibility dates between December 2020 and April 2021.

The primary outcomes for this analysis are vaccination, subjective COVID-19 mortality risk, prayer, and avoidance behavior. Vaccination is measured through a self-report of the date when the respondent received at least one dose of a COVID-19 vaccine. Subjective mortality risk is the respondent's subjective probability that he or she will die of COVID-19 within three months conditional on infection.[5] We measure prayer through an indicator of whether the respondent has prayed "to stay safe from COVID-19" in the past week.[6] We

[3]Surveyors carried out an initial survey from March 10-31, 2020 and then resurveyed respondents every two weeks through February 2021. Respondents were resurveyed every four weeks from February-June 2021. Five additional follow-ups were conducted from September 2021 through July 2023. There are 8147 unique respondents during the survey waves prior to the vaccine rollout and 11,890 unique respondents afterward. A median of 6241 respondents answer the survey in each round.

[4]Multistate, a government relations firm, created and maintained a state-by-state list of vaccine eligibility dates and criteria. The UAS elicited the occupations of respondents using six-digit O*NET occupation codes from February-April 2021. The timing of the occupation elicitation introduces the possibility that reported occupations could be endogenous to vaccine eligibility. Appendix A.6 investigates this issue and shows that vaccine eligibility is not associated with changes in employment status.

[5]The survey also measures subjective infection risk and subjective hospitalization risk. Respondents could either enter a response numerically or drag a slider from 0 to 100 percent to answer. We focus on mortality risk conditional on infection because the vaccines had the most salient impact on this margin. Estimates using other probabilities, or unconditional COVID-19 mortality risk (the product of subjective infection risk and subjective mortality risk conditional on infection), yield similar results.

[6]This measure does not include additional details about the prayer such as the duration or the frequency within the week. One concern is that highly religious people may be inframarginal for prayer over a week

create an "avoidance index" to collapse the array of measures of social distancing and the use of masks that are available in the data by computing the first principal component. Combining mask and social distancing variables into a single index is sensible because social distancing reduces the need for someone to use a mask. We normalize this index to have a mean of zero and standard deviation of one across the data.

We carry out a heterogeneity analysis according to the respondent's level of religiosity. Religiosity is measured using an index established by the Pew Research Center (Smith et al. 2015). The index includes the importance of religion in the respondent's life, attendance at religious services, prayer, and a belief in God. The prayer component is the respondent's average value of the prayer indicator described above in the survey waves before the vaccine rollout. Following the Pew approach, we divide the sample into "weakly religious", "moderately religious" and "strongly religious" subgroups. Appendix A.3 describes these variables in more detail. The religiosity components are fixed characteristics of each respondent, however belief in God was measured in 2022, and some respondents answered the importance and attendance questions only after the vaccine rollout. Appendix A.4 shows that most results, including the key prayer estimates, are similar with an index that uses only information from before the vaccine rollout.

To further explore these results, we distinguish between atheists and agnostics, who make up 26 percent of the weakly religious category, and others who identify with a denomination or describe themselves as having "no particular religion." Atheists and agnostics tend to reject spiritual practices like prayer, while others remain spiritual and sympathetic to religious practices (Lee 2014, Karim and Saroglou 2023). In our data prior to the vaccine rollout, 7 percent of weakly religious respondents who are not atheists or agnostics report praying about COVID-19, more than double the rate of 3 percent among atheists and agnostics.

Our analysis also examines heterogeneity according to the perceived effectiveness of vaccines. This variable is measured on a four-point scale in the survey wave fielded from December 9, 2020 to January 4, 2021, which precedes the respondent's own eligibility date

but could adjust prayer frequency over smaller intervals. People who pray once per week in the status quo are the most marginal for detectable effects according to our measure. In fact, supplementary data show a higher frequency of these people in the moderately and strongly religious groups than in the weakly religious group. We discuss this issue further in Section 5.2.

for 89 percent of respondents. To simplify the exposition, and because few people say that vaccines are highly ineffective, we collapse these categories and distinguish between people who perceive that vaccines are effective and people who do not.

2.3 Sample Characteristics

Table 1 provides summary statistics for the sample over the period from March-December 2020, prior to authorization and distribution of COVID-19 vaccines. Column 1 shows results for the full sample. With weighting, the sample matches the distribution of age, gender, marital status, and geographic location of the US. Over half of respondents have a chronic health condition that was used to determine vaccine eligibility.[7]

Columns 2-4 of Table 1 show how demographic and religious characteristics vary according to levels of religiosity. Although many differences are statistically significant, most comparisons are small in magnitude. Religiosity is correlated with female gender, marital status, and residence in the South. The most strongly religious subgroup in Column 4 is disproportionately Protestant and evangelical according to Table A.5.

Figure 3 illustrates the perceived effectiveness of avoidance (masks and social distancing), vaccination and prayer, distinguishing between Pew religiosity categories. People have high confidence in masks, social distancing, and vaccination as ways to prevent COVID-19, and these perceptions are very similar regardless of religiosity. By contrast, the perceived effectiveness of prayer varies sharply by religiosity. Among the least religious respondents, 15 percent believe that prayer is somewhat or very effective as a way to prevent COVID-19. 93 percent of the most religious respondents believe that prayer is somewhat or very effective, with the vast majority of these responses indicating very effective. For this group, prayer is perceived as the most effective of the four protective measures.

Panel B of Figure 3 gauges perceptions of a complementarity between prayer and science-based health inputs. We elicit whether people perceive that prayer enhances the effectiveness of either vaccines or masks, as it would if these inputs were seen as complements. Most people

[7]Table A.5 summarizes the religious denominations within the sample. Protestants are 35 percent of respondents, of whom 17 percent are evangelical; Catholics are 18 percent and other religious denominations are 18 percent. 29 percent of respondents identify as atheist or agnostic (8 percent) or report "no particular religion" (21 percent).

do not perceive complementarities, regardless of their level of religiosity. Among the least-religious cohort, 5 percent perceive that prayer makes these behaviors more effective. Even among the most-religious respondents, only 20 percent of respondents perceive that prayer enhances the effectiveness of masks or vaccines. The lack of perceived complementarity informs our model below by suggesting a large elasticity of substitution across prayer and other health inputs.[8]

3 Model

This section develops a simple model to explore the implications of the belief that prayer is a health input. We interpret the production function as describing *perceived* health production rather than objective biomedical health. For our purposes, perceived health is a person's subjective assessment of safety from COVID-19 illness and mortality, rather than psychological wellbeing or reassurance. People maximize utility over perceived health, h, and other consumption, c. They produce health by purchasing two health inputs, v and x, representing vaccination and another input, which could be either prayer or avoidance. p_v and p_x are the input prices, and we normalize the price of consumption to 1. People choose the cost-minimizing combination of vaccination and the alternative input needed to achieve their desired level of health. To proceed, we solve the cost minimization problem over x and v to derive the implicit price of health, p_h, and then maximize utility over consumption and health. We obtain theoretical predictions about the effect of vaccine eligibility by considering a discrete drop in p_v.

3.1 Health Production

People produce health using a CES production function with two inputs, v and x. α indicates the weight that the person places on x and is related to religiosity and perceived vaccine effectiveness below. The elasticity of substitution, σ, ranges from 0, which represents perfect

[8] Appendix A.2 describes a supplementary analysis of this issue. From December 2024 to February 2025, we elicited the probability that hypothetical people would contract COVID-19. These people differ in terms of whether they pray regularly, get vaccinated, and engage in avoidance behavior. Prayer and avoidance only modestly reduced perceived infection risk conditional on vaccination, further suggesting that these inputs are not seen as complements.

complements, to infinity, which represents perfect substitutes. For a given level of health, $\overline{h}$, people optimize over x and v by solving the following cost minimization problem.

$$\min_{x,v} \; p_v v + p_x x$$

$$\text{subject to}$$

$$[\alpha x^{\frac{\sigma-1}{\sigma}} + (1-\alpha) v^{\frac{\sigma-1}{\sigma}}]^{\frac{\sigma}{\sigma-1}} \;\geq\; \overline{h}$$

Solving this cost minimization problem yields the "price of health", p_h, which is the minimized cost of producing one unit of health. The conditional factor demands can be conveniently expressed in terms of p_h.

$$p_h \;=\; [\alpha^\sigma p_x^{1-\sigma} + (1-\alpha)^\sigma p_v^{1-\sigma}]^{\frac{1}{1-\sigma}} \tag{1}$$

$$v \;=\; \overline{h}\,(1-\alpha)^\sigma\, p_v^{-\sigma}\, p_h^\sigma \tag{2}$$

$$x \;=\; \overline{h}\,\alpha^\sigma\, p_x^{-\sigma}\, p_h^\sigma \tag{3}$$

Each input is proportional to the level of health and decreasing in its own price relative to the price of health. By Shephard's lemma, the elasticity of the price of health with respect to the price of vaccination equals the vaccine cost share, s_v. Since this share is positive, the price of health increases in the price of vaccination.

$$s_v \;\equiv\; \frac{\partial \ln p_h}{\partial \ln p_v} \;=\; \frac{(1-\alpha)^\sigma p_v^{1-\sigma}}{\alpha^\sigma p_x^{1-\sigma} \;+\; (1-\alpha)^\sigma p_v^{1-\sigma}} \in (0,1) \tag{4}$$

Next we maximize utility over c and h using the expression for p_h from Equation (1).

$$\max_{h,c} \; [\beta h^{\frac{\gamma-1}{\gamma}} + (1-\beta) c^{\frac{\gamma-1}{\gamma}}]^{\frac{\gamma}{\gamma-1}}$$

$$\text{subject to}$$

$$p_h h + c \leq m$$

Here β is the weight on health in utility, γ is the elasticity of substitution between health

and consumption, and m is income. Solving this utility maximization problem leads to the optimal level of health, which can be substituted into Equations (2) and (3) to obtain closed-form solutions for the health inputs.

$$h^* \;=\; \frac{m\beta^\gamma}{(1-\beta)^\gamma p_h^\gamma \;+\; \beta^\gamma p_h} \tag{5}$$

3.2 The Impact of Vaccine Eligibility

To model the arrival of the COVID-19 vaccine, we compare h^*, x^*, and v^* in the pre-vaccine period (Period 0), when $p_{v0} = \infty$, and the post-vaccine period (Period 1), when $p_{v1} > 0$ takes a small finite value. To simplify the discussion, we assume that $\sigma > 1$, so that x and v are sufficiently substitutable, consistent with the evidence presented earlier. We also assume that $\gamma \leq \sigma$, so that health and consumption are (weakly) less substitutable in utility than the health inputs are in production. This statement holds automatically if $\gamma \leq 1$, so that health and consumption are utility complements.

We consider the impact of the introduction of vaccines on v^*, h^* and x^*. For health and the alternative good, it is convenient to analyze the ratios between Period 1 and Period 0, $\frac{h_1}{h_0}$ and $\frac{x_1}{x_0}$. However, since $v_0^* = 0$, the ratio $\frac{v_1}{v_0}$ is not defined. Instead, for vaccines, we consider $\Delta v = v_1 > 0$. Evaluating Equation (2) at the Period 1 prices and the optimal level of health gives:

$$\Delta v \;=\; v_1^* \;=\; h_1^* \, (1-\alpha)^\sigma \, p_{v1}^{-\sigma} \, p_{h1}^{\sigma} \;>\; 0. \tag{6}$$

Equation (1) shows that the introduction of vaccines, which reduces p_v, also necessarily reduces p_h. Substituting p_{h0} and p_{h1} into Equation (5) and taking the ratio, we obtain the expression:

$$\frac{h_1^*}{h_0^*} \;=\; \frac{p_{h0} + (\frac{1-\beta}{\beta})^\gamma p_{h0}^\gamma}{p_{h1} + (\frac{1-\beta}{\beta})^\gamma p_{h1}^\gamma} \;>\; 1 \tag{7}$$

This expression shows that the introduction of vaccines improves perceived health.

The effect on x is most clearly seen by taking the ratio $\frac{x_1^*}{x_0^*}$. Substituting the price of health

from Equation (1) into the conditional factor demand in Equation (3) gives $x = h\,\alpha^{\sigma} p_x^{-\sigma} p_h^{\sigma}$, so that:

$$\frac{x_1^*}{x_0^*} \;=\; \underbrace{\frac{h_1^*}{h_0^*}}_{>1} \;\times\; \underbrace{\left(\frac{p_{h1}}{p_{h0}}\right)^{\sigma}}_{<1} \;<\; 1. \tag{8}$$

Equation (8) shows that the effect of vaccine availability on x reflects two opposing forces. The first term is a scale effect: as the price of health falls, the optimal level of health increases, and producing more health requires more of every input at a given input mix. The second term is a substitution effect: at the new prices, the cost-minimizing way to produce a given level of health uses less x and more v. Behavioral compensation (a reduction in x) occurs when the substitution effect is larger. Which force dominates depends on the two elasticities of substitution. The substitution effect is larger when x and v are more readily substituted in producing health (higher σ), and the scale effect is larger when people are more willing to give up consumption for health as health becomes cheaper (higher γ). Under the assumptions that $\sigma > 1$ and $\gamma \leq \sigma$, people are more willing to replace x with v than to replace c with h, so the substitution effect dominates and x falls.

This equation also shows that the change in x need not track changes in h. The substitution term operates at a given level of health, so x can decline even holding h fixed, while the change in h depends on the scale effect alone. Input utilization may therefore adjust even when perceived health changes only modestly. This prediction diverges from the coping mechanism, under which x varies with perceived risk and the resulting psychological distress, but does not depend directly on p_v.

3.3 Predictions by Perceived Vaccine Effectiveness and Religiosity

Next we consider how the effects of vaccine eligibility vary with α, the weight on the alternative good in the health production function. The α parameter captures the perceived relative productivity of x compared to v. Considering x as prayer, a large value of α is associated with both high religiosity and low confidence in vaccines. This symmetry means that predictions for religiosity and perceived vaccine effectiveness are isomorphic but oppositely

signed.

Figure 3 provides guidance about realistic magnitudes of α. No group considers prayer to be much more effective than vaccination. At most, strongly religious people perceive prayer to be moderately more effective. All religiosity groups see avoidance as slightly more effective than vaccination. These patterns are consistent with moderate and low values of α.

To obtain the differential effect on vaccination, we log-differentiate Equation (6) with respect to α:

$$\frac{d \ln \Delta v}{d\alpha} \;=\; \frac{1}{h_1^*} \frac{dh_1^*}{d\alpha} \;+\; \sigma \left[\frac{d \ln p_{h1}}{d\alpha} - \frac{1}{1-\alpha} \right]. \tag{9}$$

The first term is the effect of α on the optimal level of health, which operates through the price of health in Period 1, and the second term is the effect of α on the vaccination required per unit of health. Overall, the expression is negative for all α when $\gamma \geq 1$. It is also negative if $\gamma < 1$ unless vaccination accounts for a large share of the cost of health. We therefore expect a weaker impact on vaccination for people with a higher α, including more religious people and those who doubt the vaccine.

Next we consider the comparative static of health. Since the demand for health in Equation (5) depends only on the price of health, the health gain from the vaccine is governed by how much the vaccine lowers the price of health.

$$\frac{d}{d\alpha} \ln \left(\frac{p_{h1}}{p_{h0}} \right) \;=\; \frac{\sigma}{\sigma - 1} \frac{s_{v1}}{\alpha(1-\alpha)} \;>\; 0, \tag{10}$$

where s_{v1} is the vaccine cost share from Equation (4) in Period 1. Since Equation (10) is positive, the vaccine lowers the price of health less as α increases, which translates into a smaller health gain in:

$$\frac{d}{d\alpha} \ln \left(\frac{h_1^*}{h_0^*} \right) \;<\; 0. \tag{11}$$

We therefore expect smaller improvements in perceived health for more religious people and for people who doubt the vaccine.

Next we consider the comparative static for the alternative health input, x. Log-

differentiating Equation (8) with respect to α gives two terms of opposite sign:

$$\frac{d}{d\alpha}\ln\left(\frac{x_1^*}{x_0^*}\right) = \underbrace{\frac{d}{d\alpha}\ln\left(\frac{h_1^*}{h_0^*}\right)}_{<0} + \underbrace{\sigma\,\frac{d}{d\alpha}\ln\left(\frac{p_{h1}}{p_{h0}}\right)}_{>0}. \tag{12}$$

The first term is the scale channel: people with higher α obtain a smaller health gain from vaccination, leading to a weaker increase in the demand for health and a larger decline in x. The second term is the substitution channel: the vaccine lowers the price of health by less for people with higher α (Equation (10)), leading to less substitution away from x. The second channel formally captures the concept of inframarginality: as α approaches one, x is the only input that matters, which means that the vaccine does not lower the price of health or affect the optimal utilization of x. The second channel dominates the first whenever $\gamma \geq 1$, as well as when $\gamma < 1$ and vaccination is a sufficiently large share of the cost of health, so that $s_{v1} \geq (1-\alpha)(1-\gamma)/(\sigma-1)$. We therefore expect a smaller proportional decline in x for people with higher α. In the case of prayer, strongly religious people should reduce prayer by less because prayer is more inframarginal for them, even though their smaller health gain works in the opposite direction. Since the perceived effectiveness of avoidance does not vary with religiosity in Figure 3, we do not expect differences by religiosity in the impact of vaccination on avoidance.

Table 2 summarizes these theoretical predictions. The direct effect of the vaccine rollout should be to increase vaccination, reduce subjective mortality risk, and reduce prayer and avoidance. We expect a weaker effect on vaccination, subjective mortality risk and prayer for more religious people. However, we do not predict a difference in the impact on avoidance by religiosity since all groups have similar beliefs about the effectiveness of avoidance and vaccines in Figure 3. By a similar theoretical logic, the vaccine rollout has a larger effect for people who believe in vaccines, and these people experience larger declines in subjective mortality risk, prayer, and avoidance. Since they derive from the common mechanism of α, these sources of heterogeneity provide complementary predictions.

4 Analysis of the COVID-19 Vaccine Rollout

Next we test the model's predictions by evaluating the impact of the COVID-19 vaccine rollout on vaccination, perceived mortality risk, prayer and science-based avoidance behavior. Vaccine eligibility may affect prayer through multiple channels. Eligibility lowers the price of a substitute health input, which induces vaccination and thereby lowers perceived COVID-19 risk. The reduced-form estimate therefore combines the response to the lower price with any response to lower risk, since both follow from the same eligibility change. Section 5 uses the pattern of effects across groups and outcomes to distinguish these pathways.

We estimate the impact of vaccine eligibility using a difference-in-differences approach that compares the change in outcomes for early-eligible and late-eligible respondents. Given the staggered timing of eligibility, we implement the imputation-based difference-in-differences estimator of Borusyak et al. (2024).[9]

$$Y_{it} = \mu_i + \lambda_{te} + \delta E_{it} + \epsilon_{it} \tag{13}$$

In this expression, i indexes the individual and t indexes the survey date. e indicates time-constant categories of perceived vaccine effectiveness.[10] E_{it} is an indicator that the respondent is eligible for the vaccine on date t. We incorporate post-stratification weights to align the sample with the demographic characteristics of US adults. Standard errors are clustered by state of residence immediately prior to the vaccine rollout, consistent with state-level determinations of vaccine eligibility.[11] For the heterogeneity analyses, we estimate the same

[9] The Borusyak et al. (2024) imputation estimator is efficient in our setting of many staggered eligibility cohorts and no never-treated comparison group. The alternative Callaway and Sant'Anna (2021) estimator requires aggregating the data to discrete periods with wave-level treatment timing, discarding intra-wave variation in vaccine eligibility. Appendix Table A.4 compares the two estimators. With wave-level aggregation, neither estimator yields a precise estimate for prayer or avoidance, and the Borusyak et al. (2024) estimator reproduces our main estimates for the same respondents once the daily data structure is restored.

[10] We allow the time fixed effects to vary by perceived vaccine effectiveness because the two groups follow distinct time trends before the rollout. The interactions of time indicators with e are jointly significant for perceived mortality risk, prayer and avoidance, with $p \leq 0.008$ in every case. Under a common time trend, λ_t, the estimates for vaccination, perceived mortality risk, and prayer are essentially unchanged, while the avoidance estimate and the heterogeneity in avoidance by religiosity are not significant ($p = 0.11$ and $p = 0.33$, respectively).

[11] Clustering by state allows shocks to be correlated within a state across eligibility cohorts and over time,

model with time fixed effects that also vary by the dimension of heterogeneity, and report the average effect among treated observations in each group. We test the equality of these group-specific effects with Wald tests.

Identification requires that the eligibility date is not correlated with differential trends in the outcome variables. In this context, eligibility criteria also directly induce level differences in the characteristics that were the basis for eligibility. As we describe above, authorities first offered the vaccine to older people and members of prioritized professions. Dividing the sample at the median eligibility date (March 8, 2021), Columns 5-6 of Table 1 illustrate that, as expected, early-eligible people are older and more likely to be employed in health and education fields. Eligibility is not associated with gender and the difference in education is small. The table also shows that people with early eligibility are modestly more religious. Individual fixed effects absorb these level differences, along with other time-invariant differences across respondents.

To assess the possibility of pre-eligibility trends in the primary outcomes, Figure 4 plots the patterns in vaccination, mortality risk, prayer, and avoidance for 24 weeks before and 12 weeks after eligibility. We divide the sample into two-week periods and estimate coefficients on these periods within the Borusyak et al. (2024) framework. The pre-eligibility coefficients for vaccination and prayer are close to zero until the last two weeks before eligibility, when vaccination rises slightly and prayer dips. States announced eligibility phases before they took effect, so some respondents may have obtained access before their coded date. The prayer coefficients prior to the final period are jointly insignificant ($p = 0.16$), and Panel C of Table A.2, which drops the two weeks before eligibility, gives a similar prayer estimate. In addition, Figure A.3 plots pre-trends by religiosity. Among weakly religious respondents,

as when a wave of infections affects people who become eligible on different dates. A finer level, such as state by eligibility date, would yield more clusters but would rule out this correlation. Because there are several ways to conduct inference in this setting, we consider two alternatives to the standard errors in our tables. The first is the leave-out variance estimator of Borusyak et al. (2024), which is more conservative when eligibility cohorts are small. The second is a wild cluster bootstrap of the t-statistics from the leave-out estimator, with the null imposed and 1,000 replications (Cameron et al. 2008, MacKinnon and Webb 2017). Under these two alternatives, the p-values are 0.05 and 0.06 for the overall effect on prayer, 0.02 and 0.02 for the effect on prayer among weakly religious people, and 0.06 and 0.07 for the difference between weakly and strongly religious people. The effects on vaccination and perceived mortality risk remain significant at the 1 percent level under both alternatives. For avoidance, the overall effect has p-values of 0.05 and 0.17, and the differences by religiosity are not statistically significant under either alternative. Additional estimates are available from the authors.

who account for the prayer response, the pre-eligibility coefficients for prayer show no trend (slope of 0.00 p.p. per period, $p = 0.96$). The coefficients for avoidance drift downward before eligibility but are jointly insignificant ($p = 0.30$); the decline steepens after eligibility ($p = 0.08$ for the change in slope), and the estimate is larger when we drop the two weeks before eligibility (0.065 s.d. in Panel C of Table A.2). Perceived mortality risk declines by about 0.13 p.p. per two-week period before eligibility, so the estimate may include a continuation of the earlier decline; after eligibility it declines nearly five times as fast ($p < 0.001$ for the difference in slopes).

5 Results

5.1 Overall Estimates and Heterogeneity by Perceived Vaccine Effectiveness

Overall estimates of the impact of vaccine eligibility appear in Panel A of Table 3, and the signs of these results align with the theoretical predictions in Table 2 (Panel A). In Columns 1 and 2, vaccination rises by 13 p.p. and perceived mortality risk declines by 1.6 p.p. (8 percent).[12] In Columns 3 and 4, prayer declines by 0.9 p.p. and avoidance falls by 0.05 s.d. Scaling by the first-stage effect on vaccination yields Wald estimates of -6.8 p.p. for prayer and -0.38 s.d. for avoidance among people induced to vaccinate.[13] The negative sign of the prayer result supports the assumption of the model that prayer and vaccination are substitutable enough for the substitution effect to dominate. If the inputs were complements, we would expect vaccine eligibility to increase prayer.

Panel B of Table 3 shows estimates of heterogeneity by perceived vaccine effectiveness. As predicted, estimates are larger for people who perceive that vaccines are effective. For this group, vaccination increases by 16 p.p., subjective mortality risk declines by 2.0 p.p., prayer declines by 1.0 p.p. and avoidance falls by 0.05 s.d. Since this group makes up 82 percent of the sample, these estimates resemble the overall estimates in Panel A. By contrast, results are much smaller for people who perceive that the vaccines are ineffective, although

[12] Figure 4 shows that perceived mortality risk was already declining before eligibility. When we add a linear time trend that varies by month of eligibility, the estimate is 1.1 p.p.

[13] We should interpret these estimates cautiously because vaccine eligibility could influence these outcomes through anticipatory, informational, or social channels in addition to actual vaccination.

differences for prayer and avoidance are not statistically significant. These patterns align with the predictions in Panel B of Table 2, and suggest that effects operate through beliefs about the vaccine rather than concurrent changes such as the reopening.

Next we assess the robustness of these results. With individual fixed effects, our main specification controls for individual time-constant attributes, however differential trends that are correlated with vaccine eligibility remain a threat to validity. Figure 5 reproduces the estimates in Table 3 and then controls sequentially for the interaction of time indicators with political party indicators, age groups, census regions, and education categories.[14] In general, allowing for more flexible time trends does not substantially change the estimates.[15]

Appendix A.4 carries out additional robustness tests. Table A.2 shows that results are robust if we shift the eligibility date 14 days forward or backward. This outcome is reasonable because once someone becomes eligible, it takes time to get vaccinated and then for risk perceptions and behavior to adjust. The appendix also shows that results are robust under alternative ways of constructing perceived vaccine effectiveness and avoidance.

5.2 Heterogeneity by Religiosity

Heterogeneity estimates by religiosity appear in Table 4. Despite the notion of vaccine hesitancy among religious people and the prediction of the model, the impact on vaccination only varies slightly by religiosity in Column 1. The effect is 15 p.p. for weakly religious people, 13 p.p. for moderately religious people, and 14 p.p. for strongly religious people. In Column 2, the effect on perceived mortality risk is larger for weakly religious people, which is consistent with the model ($p = 0.01$), although the effect is similar for the moderately and strongly religious groups.

[14]Education is divided into four categories: (1) high school graduate, GED, or less; (2) some college with no degree, or an associate degree from an occupational or vocational program; (3) an associate degree from an academic program, or a bachelor's degree; and (4) a master's, professional, or doctoral degree. Observations are grouped into three age categories: 18–39, 40–59, and 60+. There are four census regions: Northeast, Midwest, South, and West. Political party affiliation is divided into four categories: (1) Democrats or Greens, (2) Republicans or Libertarians, (3) independents, members of other parties, or people not aligned with any party, and (4) respondents who did not report a party.

[15]Eligibility is mechanically determined by age, occupation, chronic health status, and state rollout timing. Fully interacting time trends with these characteristics would absorb part of the identifying variation. Instead, identification relies on parallel pre-trends across eligibility cohorts, which we assess directly in Figure 4. Appendix A.4 reports additional robustness tests.

Next we turn to prayer and avoidance. Column 3 shows that prayer falls by 1.6 p.p. (26 percent) for weakly religious people, by 1.2 p.p. (2 percent) for moderately religious people, and by 0.1 p.p. (0.1 percent) for strongly religious people ($p = 0.01$ for the strong-weak difference).[16] This pattern is consistent with inframarginality in the model: the vaccine lowers the price of health less for strongly religious respondents and so they substitute away from prayer less. In Column 4, less religious people also have larger effects on avoidance, although this gradient is smaller than the gradient for prayer and is not robust under the alternative inference in footnote 11 or to the specification in footnote 10.[17,18]

One concern is that weekly prayer is measured as a binary indicator, which could obscure within-week adjustments among more religious people. Using supplementary data on the frequency of prayer within the week, Figure 6 shows that moderately and strongly religious people have substantial mass at one instance per week, indicating that many remain marginal at the weekly level. 16 percent of moderately religious people and 8 percent of strongly religious people pray once per week, compared to 7 percent of weakly religious people. This evidence suggests that the lack of a prayer response among strongly religious people is unlikely to be an artifact of the binary measurement of prayer.

Distinguishing between atheists/agnostics and others can help us understand the strong prayer response for the weakly religious subgroup. Figure 7 reports estimates for atheists and agnostics and for all other respondents, in the full sample and within the weakly religious category. While vaccination rises and perceived mortality risk falls for both groups, atheists and agnostics have larger impacts on both outcomes ($p < 0.001$ for these comparisons).

[16]The religiosity index includes the respondent's average pre-rollout response to the prayer measure that is also our outcome. To confirm that this component does not drive the pattern, we construct the index from the other three components. With this index, the estimates for prayer are -2.4 p.p., -1.0 p.p. and 0.4 p.p. for weakly, moderately and strongly religious people ($p < 0.001$ for equality across the three groups). The two classifications agree for 85 percent of observations, and the gradient in avoidance is not significant with the alternative index ($p = 0.49$).

[17]Since we measure the perceived effectiveness of prayer, it is also possible to estimate heterogeneity along this margin. In practice, religiosity and perceived prayer effectiveness are highly correlated. 15 percent of weakly religious people think prayer is effective and 93 percent of strongly religious people hold this belief. Analyzing heterogeneity on this margin instead of religiosity leads to similar results. People who believe that prayer is ineffective show larger absolute impacts on vaccination ($p = 0.03$) and mortality risk ($p < 0.001$). People who believe that prayer is ineffective have larger point estimates for prayer and avoidance but the differences are not statistically significant ($p = 0.13$ for prayer and $p = 0.44$ for avoidance).

[18]Figure A.9 shows heterogeneity by religiosity after controlling sequentially for the interaction of time indicators with several sets of controls, as in Figure 5. Estimates are robust with these controls, suggesting that heterogeneity by religiosity is not simply an artifact of these factors.

Atheists and agnostics do not reduce prayer, which is near zero before the vaccine rollout, but sharply reduce avoidance. By contrast, other weakly religious respondents reduce prayer by 1.8 p.p. (26 percent) but have a weaker and statistically insignificant effect on avoidance. Within the weakly religious subgroup, the difference between atheists/agnostics and others is statistically significant for prayer ($p = 0.03$) but not for avoidance ($p = 0.11$). The relatively large impact on prayer coupled with a relatively small impact on perceived mortality risk for other weakly religious respondents is difficult to explain with coping, which would predict alignment between the magnitudes of these effects. By contrast, Equation (8) of the model shows that the prayer response also reflects the marginality of this input. In summary, both the prayer response and (to a lesser degree) the avoidance response are concentrated among the subgroups for whom these inputs are marginal.

5.3 Interpretation

The effect of vaccine eligibility on prayer is a reduced-form result that could arise through alternative channels. In Section 3, the invention of vaccines reduces the relative price of this input from infinity to a small positive value, which leads people to substitute out of prayer. Another interpretation is that the vaccine rollout led to the reopening of public spaces, which increased the opportunity cost of prayer and thereby caused the observed effects. A third possibility is coping, in which prayer responds to distress and falls as perceived risk declines.

Table 5 reproduces our main estimates for additional outcomes that allow us to explore these possibilities. First we consider the reopening channel. The reopening of churches might have reduced the reliance on private prayer. Column 1 of the table shows results for in-person worship, which is measured through a binary indicator of weekly participation, in the same way as prayer. Attendance rises by 0.45 p.p., half the size of the decline in prayer, and the pattern by religiosity does not match the pattern for prayer, suggesting that a shift toward in-person worship is unlikely to account for the decline in prayer. A more general version of the concern is that the rollout expanded the activities available outside the home, mechanically reallocating time away from home-based activities such as prayer. This mechanism is not especially plausible because prayer, which takes little time, is unlikely to be displaced by the

reopening of time-intensive activities.[19] Moreover, prayer is not necessarily tied to the home: a recent national survey showed that US adults often pray in their cars (Skylight 2023), and so spending less time at home need not affect prayer. Column 2 estimates the impacts on an index of activities outside the home. We find a small and insignificant overall effect in Panel A and results by religiosity in Panel C that do not align with the prayer results, further casting doubt on this possibility. While this result might seem at odds with the significant negative effect on avoidance in Table 3, an analysis of the components of the avoidance index in Figure A.4 shows that a change in mask usage rather than social distancing mainly drives that result.

Coping is another possible interpretation of the effect on prayer. According to the literature on the psychology of religion, prayer functions as a coping resource that helps people manage threats that they cannot face on their own, helping to reduce anxiety and restore a sense of control (Pargament 2001, Ano and Vasconcelles 2005, Upenieks 2023). Economists have interpreted the increase in prayer during the pandemic and during natural disasters in these terms (Bentzen 2019, Bentzen 2021). Under this interpretation, vaccine eligibility may reduce prayer simply by reducing COVID-19 risk, irrespective of any perceived protective effect of prayer. A decline in prayer should be accompanied by improvements in anxiety or a sense of control under this explanation. Columns 3 and 4 of Table 5 show estimates for the GAD-2 anxiety scale and the "lack of control" factor of the Perceived Stress Scale (Kroenke et al. 2007, Ingram et al. 2016). Vaccine eligibility does not reduce either outcome in the overall sample. It only reduces lack of control in the strongly religious subgroup, for whom prayer does not change. There is a small *increase* in anxiety among weakly religious people, which is the group with the largest declines in prayer. In sum, these results do not support the coping channel.[20]

[19] In a survey from December 2024 to February 2025 (described in Appendix A.1), prayers last a median of 3 minutes.

[20] The absence of an effect on anxiety differs from studies that find that COVID-19 vaccination reduced psychological distress, including Perez-Arce et al. (2021) and Agrawal et al. (2021). These studies estimate the effect of receiving a vaccine dose rather than the effect of eligibility. Since eligibility raises vaccination by 13 p.p. in Table 3, the Perez-Arce et al. (2021) estimate implies an effect of 0.005 s.d. per eligible person, which is within our confidence interval. Agrawal et al. (2021) report a 9.4 p.p. reduction in the share of adults with anxiety symptoms (GAD-2 $\geq$ 3). Even taking this estimate as given, a decline in anxiety cannot explain our prayer estimates because there is at most a 2.5 p.p. difference in the frequency of prayer between high-anxiety and low-anxiety individuals in our data.

In summary, we observe an impact of vaccine eligibility on prayer without accompanying improvements in anxiety or perceived control. The prayer effect is concentrated among weakly religious people, whereas the literature attributes religious coping primarily to strongly religious people (Pargament 2001, Bentzen 2021). While distress may also lead people to pray, these patterns support the health production interpretation of our results.

6 Discussion

This paper provides evidence that religious practice can function as a perceived health input. People pray less once they become eligible for the vaccine, as the model predicts when the price of a substitute input falls. The heterogeneity in this response clarifies the economic structure underlying religious practice. Among weakly religious people, and particularly those who are not atheists or agnostics, prayer behaves as a discretionary and situational input. For strongly religious people, prayer is unresponsive to vaccine eligibility. This pattern is consistent with inframarginality: inputs that are central to identity or daily routine remain stable when relative prices change, while marginal inputs adjust.

These findings suggest that we can apply economic reasoning to culturally embedded practices. When modeled as perceived inputs into health production, religious practices can exhibit standard marginal substitution patterns. More generally, practices that are deeply integrated into identity may be inframarginal and therefore resistant to economic incentives. We interpret the decline in prayer as substitution across perceived health inputs. An important alternative is coping, in which prayer responds to anxiety and loss of control. Two results suggest that coping alone does not explain our findings. First, vaccine eligibility does not reduce anxiety or improve the sense of control, as it should under this explanation. Second, the prayer effect is concentrated among weakly religious people while the literature on religious coping argues that strongly religious people exhibit this behavior the most. Results are more consistent with a response to a change in relative prices.

We do not find evidence that the effect on vaccination varies by religiosity. The impact on vaccination is nearly the same across religiosity groups, which challenges the common conception of vaccine hesitancy among religious people.

It may be overly narrow to represent the centrality of prayer solely through its weight in the health production function. For many religious people, prayer is a method to reinforce a relationship with God that in turn guides their daily decisions. In this case, protection against a particular disease may be an incidental reason to pray. In general, religiosity could enter the optimization problem through several other parameters, including preferences, prices and the elasticity of substitution among other inputs.

More broadly, the results underscore the importance of subjective production technologies. Individuals act on their beliefs about how actions translate into outcomes and adjust behavior when objective risk or relative prices change. Similar dynamics may arise outside of health. For instance, superstitions may influence financial portfolio choices and beliefs about predestination may shape investment in child human capital. Behavior based on supernatural beliefs can be analyzed as rational (Leeson 2012, Leeson 2014). Incorporating such behaviors into a subjective production framework allows us to analyze their responsiveness to prices using standard economic tools.

Table 1: Pre-Rollout Sample Characteristics by Religiosity and Vaccine Eligibility

		Religiosity			Vaccine Eligibility		P-values	
	All	Weak	Moderate	Strong	Early	Late	Relig.	Elig.
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
A: Demographics								
Age	49.8	45.6	50.1	52.4	58.0	41.6	0.00	0.00
Female	0.51	0.44	0.51	0.57	0.51	0.51	0.00	0.77
Married	0.56	0.50	0.54	0.62	0.59	0.53	0.00	0.00
Education (16-point scale)	10.9	11.3	10.8	10.7	11.0	10.7	0.00	0.01
Employed	0.53	0.55	0.55	0.50	0.43	0.65	0.01	0.00
Healthcare/education field	0.11	0.10	0.10	0.12	0.18	0.03	0.13	0.00
Republican	0.27	0.15	0.26	0.38	0.31	0.23	0.00	0.00
Northeast census region	0.18	0.20	0.19	0.14	0.23	0.13	0.07	0.04
Midwest census region	0.21	0.24	0.21	0.19	0.22	0.19	0.31	0.53
South census region	0.38	0.27	0.36	0.48	0.36	0.40	0.00	0.59
West census region	0.23	0.29	0.23	0.20	0.19	0.28	0.04	0.21
B: Health and Religion								
Chronic health condition	0.55	0.52	0.54	0.59	0.75	0.35	0.00	0.00
Subjective COVID mortality risk	0.19	0.17	0.20	0.20	0.23	0.16	0.00	0.00
COVID infection (last 2 weeks)	0.01	0.02	0.01	0.01	0.01	0.02	0.68	0.02
Perceived vaccine effectiveness	2.82	2.88	2.82	2.78	2.86	2.78	0.00	0.00
Perceived avoidance effectiveness	3.51	3.55	3.52	3.47	3.55	3.46	0.07	0.00
Perceived prayer effectiveness	2.83	1.53	2.88	3.58	2.88	2.77	0.00	0.05
Pew Religiosity Index	0.40	-2.89	0.17	2.95	0.59	0.21	0.00	0.00
Prays weekly about COVID	0.60	0.06	0.61	0.94	0.63	0.56	0.00	0.00
Avoidance index	-0.03	-0.00	-0.02	-0.06	0.02	-0.08	0.23	0.00
Number of Observations	139,400	33,351	60,911	45,138	66,107	73,243	–	–

Note: The table reports sample means for the period from March 11-December 10, 2020. All statistics use post-stratification weights to ensure that the sample is representative of US adults in terms of gender, race, age, marital status, and education, as explained in the text. Vaccine eligibility categories are split at the median of March 8, 2021. P-values test equality across religiosity categories (column 7) and across eligibility groups (column 8), using standard errors clustered at the state of eligibility level.

Table 2: Theoretical Sign Predictions for Vaccine Eligibility Effects

	Outcome				
	Vaccination	Subjective Mort. Risk	Prayer	Avoidance	Conditions for Signed Effect
	(1)	(2)	(3)	(4)	(5)
A: Direct effect	+	−	−	−	Vaccination and mortality risk: none. Prayer and avoidance: $\sigma > 1$ and $\gamma \leq \sigma$.
B: Vaccine effectiveness interaction	+	−	−	−	Vaccination: $\sigma > 1$ and $\gamma \leq \sigma$; if $\gamma < 1$, also $s_{v1} \leq (\sigma - 1)/(\sigma - \gamma)$. Mortality risk: $\sigma > 1$. Prayer and avoidance: $\sigma > 1$ and $\gamma \leq \sigma$; if $\gamma < 1$, also $s_{v1} \geq (1 - \alpha)(1 - \gamma)/(\sigma - 1)$.
C: Religiosity interaction	−	+	+	0	Vaccination, mortality risk, and prayer: as in Panel B. Avoidance: perceived effectiveness of avoidance does not vary with religiosity.

Note: This table summarizes the sign predictions derived in Sections 3.2 and 3.3. Panel B reports the sign of the difference in the effect between people who perceive the vaccine as effective and those who do not (lower versus higher α). Panel C reports the sign of the difference between more and less religious people (higher versus lower α). Column (5) lists sufficient conditions for the signs shown; s_{v1} is the vaccine cost share in Period 1 from Equation (4).

Table 3: The Impact of Vaccine Eligibility

	Vaccination	Mortality Risk	Prays Weekly	Avoidance Index
	(1)	(2)	(3)	(4)
A: Overall Impact				
Overall	0.13***	-0.016***	-0.0091***	-0.052**
	(0.0073)	(0.0013)	(0.0034)	(0.025)
Pre-rollout mean	0.00	0.19	0.60	-0.03
Observations	183,534	183,534	183,534	164,263
B: Impact by Perceived Vaccine Effectiveness				
Vaccine not effective	0.027***	0.0019	0.0043	-0.022
	(0.0054)	(0.0032)	(0.011)	(0.045)
Vaccine effective	0.16***	-0.020***	-0.0099***	-0.054**
	(0.0086)	(0.0014)	(0.0034)	(0.024)
Pre-rollout mean				
Vaccine not effective	0.00	0.17	0.70	-0.34
Vaccine effective	0.00	0.20	0.58	0.06
Equality of coefficients (p-value)	0.00	0.00	0.22	0.30
Observations	183,534	183,534	183,534	164,263

Note: Regressions are based on the specification: $Y_{it} = \mu_i + \lambda_{te} + \delta E_{it} + \epsilon_{it}$ in which i indexes the individual, t indexes the survey date, e is perceived vaccine effectiveness, and E_{it} is the individual's vaccine eligibility. Estimates follow the difference-in-difference imputation approach of Borusyak et al. (2024). Estimates are weighted to be nationally representative in terms of sex, race, age, marital status, and education, as explained in the text. Standard errors are clustered at the state of eligibility level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: The Impact of Vaccine Eligibility by Religiosity

	Vaccination	Mortality Risk	Prays Weekly	Avoidance Index
	(1)	(2)	(3)	(4)
Weakly religious	0.15***	-0.023***	-0.016***	-0.070***
	(0.0080)	(0.0024)	(0.0031)	(0.023)
Moderately religious	0.13***	-0.014***	-0.012**	-0.044**
	(0.012)	(0.0019)	(0.0059)	(0.022)
Strongly religious	0.14***	-0.015***	-0.0013	-0.029
	(0.0087)	(0.0017)	(0.0042)	(0.030)
Pre-rollout mean				
Weakly religious	0.00	0.16	0.06	0.00
Moderately religious	0.00	0.19	0.61	-0.02
Strongly religious	0.00	0.20	0.94	-0.05
Equality of coefficients (p-value)	0.42	0.01	0.03	0.05
Weak vs. strong (p-value)	0.43	0.01	0.01	0.02
Observations	183,477	183,477	183,477	164,215

Note: Regressions are based on the specification: $Y_{it} = \mu_i + \lambda_{te} + \delta E_{it} + \epsilon_{it}$ in which i indexes the individual, t indexes the survey date, e is perceived vaccine effectiveness, and E_{it} is the individual's vaccine eligibility. Estimates also include the interaction between time indicators and religiosity categories. Estimates follow the difference-in-difference imputation approach of Borusyak et al. (2024). Estimates are weighted to be nationally representative in terms of sex, race, age, marital status, and education, as explained in the text. Standard errors are clustered at the state of eligibility level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 5: Alternative Explanations for the Decline in Prayer

	Attends Rel. Services	Going Outside	Anxiety (GAD-2)	Lack of Ctrl (PSS-4)
	(1)	(2)	(3)	(4)
A: Overall Impact				
Overall	0.0045**	0.014	0.0049	0.014*
	(0.0022)	(0.0091)	(0.0076)	(0.0073)
B: Impact by Perceived Vaccine Effectiveness				
Vaccine not effective	-0.0039	-0.022	0.017	0.029
	(0.0079)	(0.023)	(0.014)	(0.021)
Vaccine effective	0.0060**	0.019*	0.0040	0.014
	(0.0024)	(0.011)	(0.0095)	(0.0090)
Equality of coefficients (p-value)	0.23	0.09	0.46	0.57
C: Impact by Religiosity				
Weakly religious	0.0006	0.018	0.030**	0.022
	(0.0016)	(0.017)	(0.015)	(0.017)
Moderately religious	0.0050	0.025**	0.0059	0.038***
	(0.0031)	(0.011)	(0.013)	(0.010)
Strongly religious	-0.0008	-0.012	-0.0094	-0.023**
	(0.0054)	(0.014)	(0.012)	(0.011)
Equality of coefficients (p-value)	0.33	0.03	0.10	0.00
Weak vs. strong (p-value)	0.79	0.16	0.04	0.04

Note: Regressions are based on the specification: $Y_{it} = \mu_i + \lambda_{te} + \delta E_{it} + \epsilon_{it}$ in which i indexes the individual, t indexes the survey date, e is perceived vaccine effectiveness, and E_{it} is the individual's vaccine eligibility. In Panel C, time indicators are additionally interacted with religiosity category, so that differential effects by religiosity are not estimated under an imposed common time path. Attendance at religious services is an indicator for attending an in-person religious service in the past week. Going outside includes having attended a gathering with more than 10 people, gone out to a bar, club, or other place where people gather, gone to the grocery store or pharmacy, gone to a friend, neighbor, or relative's residence, gone outside to walk, hike, or exercise, traveled by airplane, and traveled by public transportation; we compute the first principal component of the included variables and then standardize this index. Column 3 is the two-item generalized anxiety subscale (GAD-2) of the PHQ-4 (Kroenke et al. 2009). Column 4 is the lack-of-control component of the four-item Perceived Stress Scale (Cohen et al. 1994), comprising the two items asking how often the respondent has felt unable to control the important things in life and that difficulties were piling up so high that they could not be overcome. Columns 3 and 4 are scored so that higher values indicate worse mental health and are standardized to have mean zero and unit standard deviation. Estimates follow the difference-in-difference imputation approach of Borusyak et al. (2024). Estimates are weighted to be nationally representative in terms of sex, race, age, marital status, and education, as explained in the text. Standard errors are clustered at the state of eligibility level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

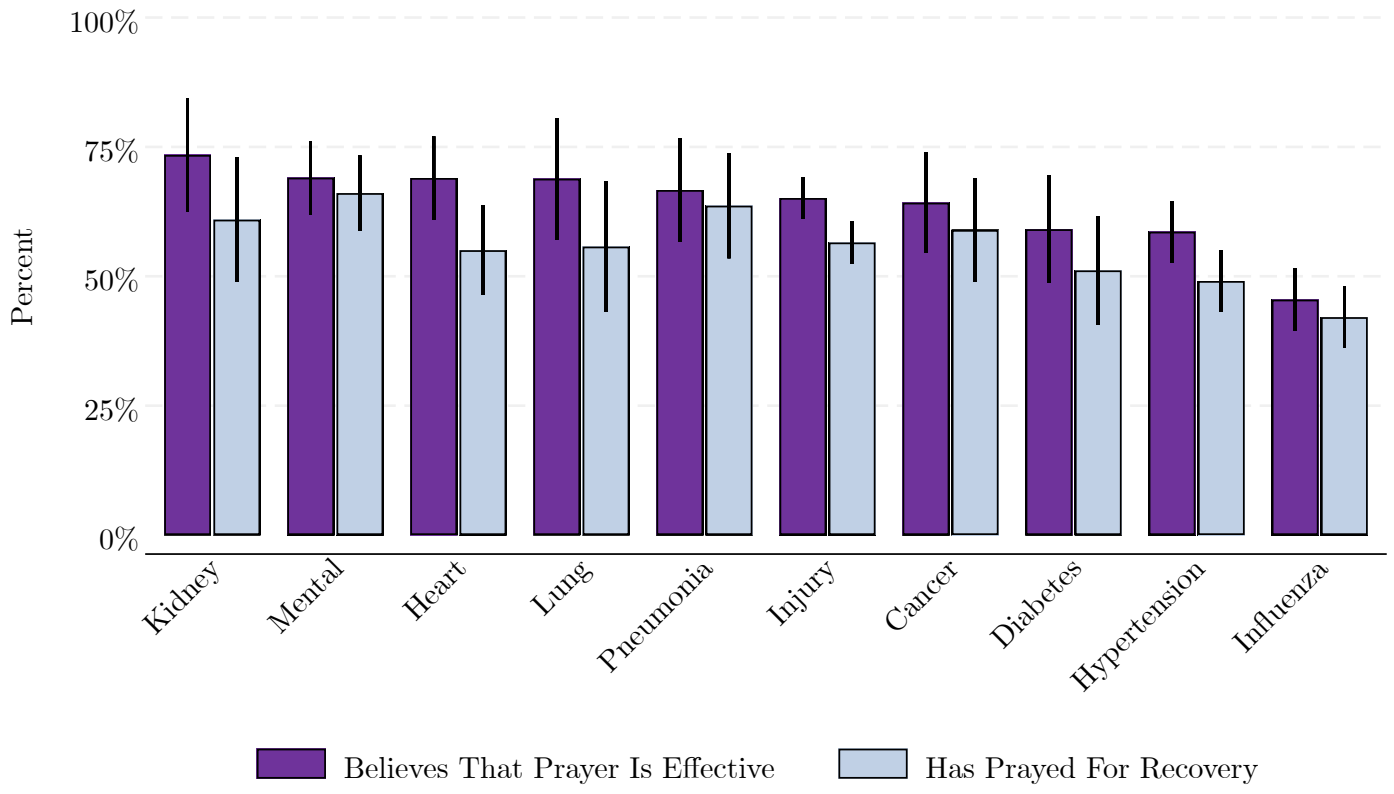


Figure 1: Perceived Effectiveness and Utilization of Prayer across 10 Health Problems

Note: The figure shows perceived effectiveness and utilization of prayer across 10 health problems. Error bars show 95 percent confidence intervals. Purple bars report the percentage of respondents who perceive prayer as an effective way to recover from health problems, while the blue bars report the percentage of respondents who have prayed as a way to recover from health problems. The sample is based on a survey of 1814 participants in the Understanding America Study that was conducted from December 17, 2024 to February 9, 2025.

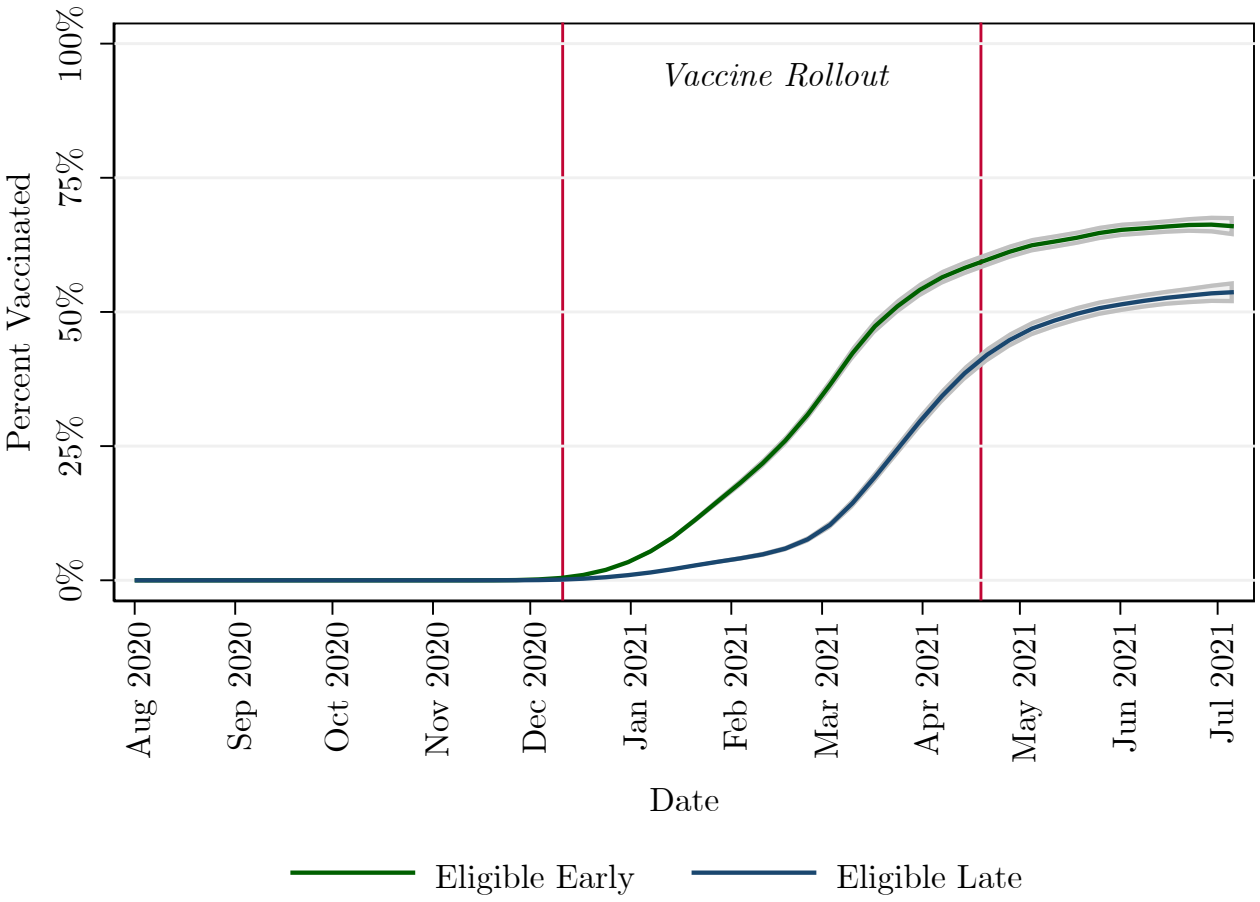
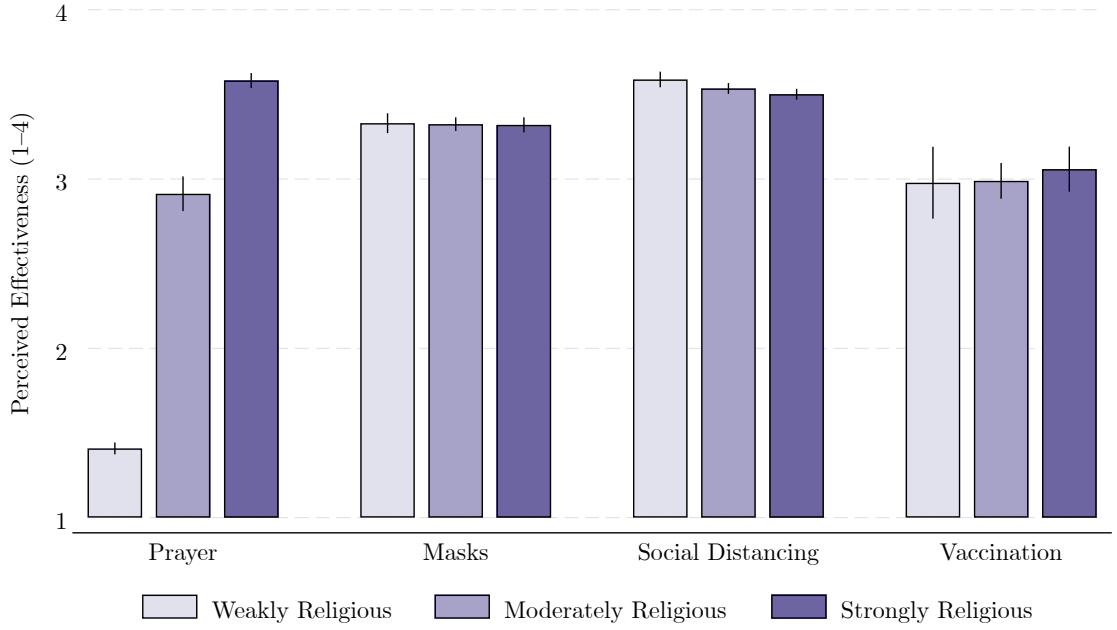


Figure 2: Vaccine Eligibility and the Timing of the Vaccine Rollout

Note: The figure divides the sample according to the median date of vaccine eligibility. Error bars show 95 percent confidence intervals based on state-clustered standard errors. Estimates are weighted to be nationally representative of US adults, as explained in the text.



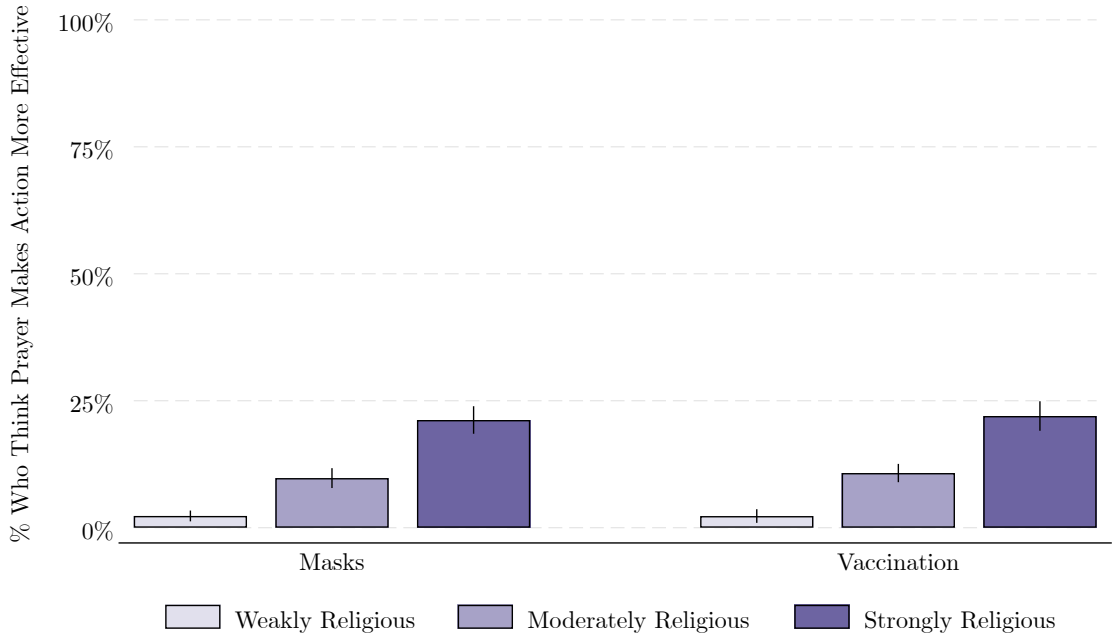


Figure 3: Perceptions About the Effectiveness and Complementarity Between COVID-19 Prevention Behaviors

Note: The figures divide the sample into religiosity categories. Error bars show 95 percent confidence intervals. Estimates are weighted to be representative of US adults, as described in the text. Panel A reports the average perceived effectiveness of prayer, masks, social distancing, and vaccination against COVID-19 on a four-point Likert scale. Panel B shows the percent of respondents who perceive a complementarity between prayer and other health inputs.

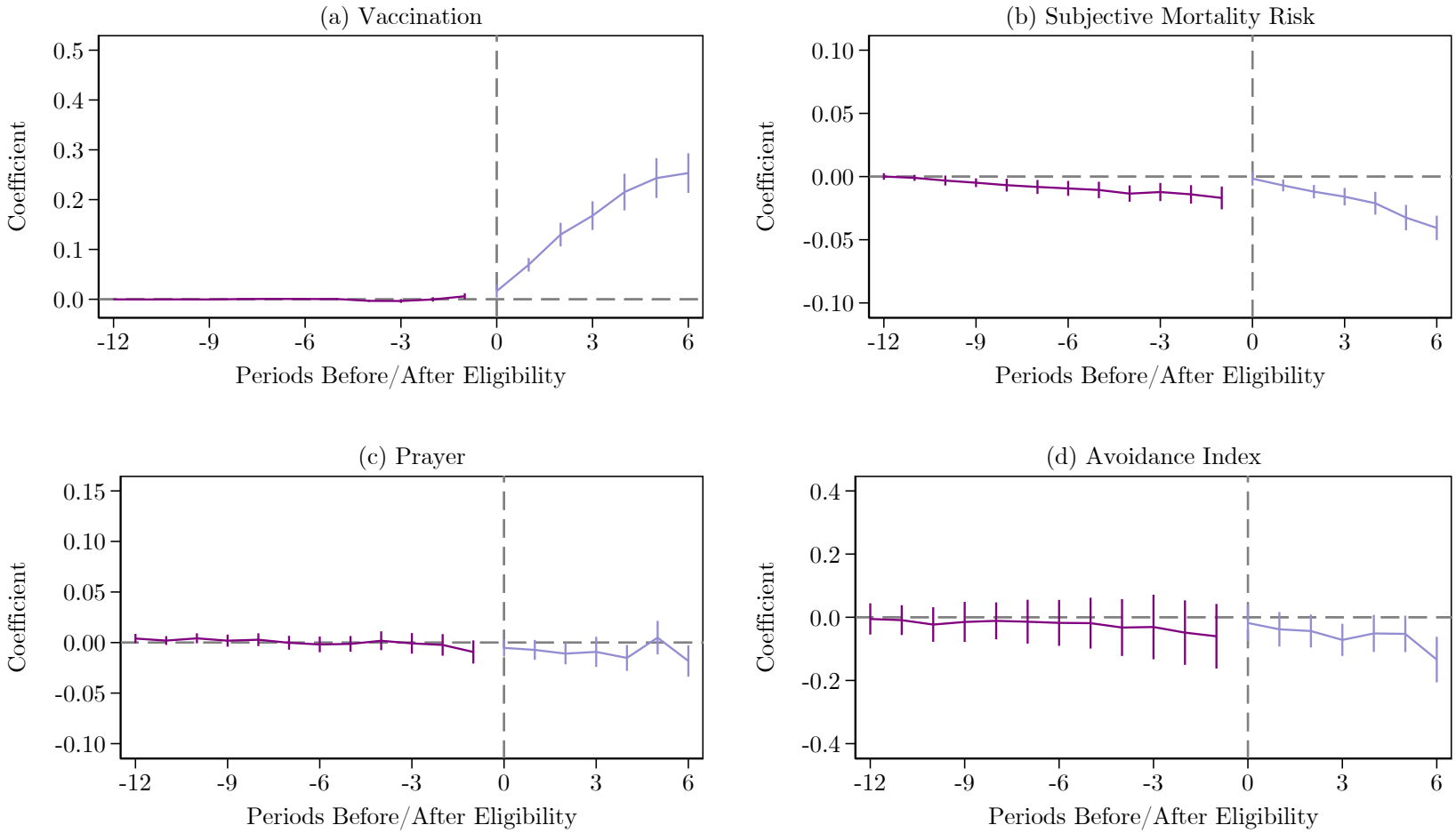


Figure 4: Pre-Trends in Key Outcomes

Note: The figure estimates biweekly pre- and post-eligibility trends for key outcome variables. Estimates are based on the DID imputation method of Borusyak et al. (2024). Variables are defined as in the text. Periods are two weeks long. Error bars indicate 95 percent confidence intervals based on state-clustered standard errors. Estimates are weighted to be nationally representative. Pre-eligibility coefficients are measured relative to periods more than 24 weeks before eligibility, while post-eligibility coefficients are measured relative to the imputed counterfactual, so the two series do not share a baseline. The post-eligibility estimates assume parallel trends and do not net out any pre-existing differential trend.

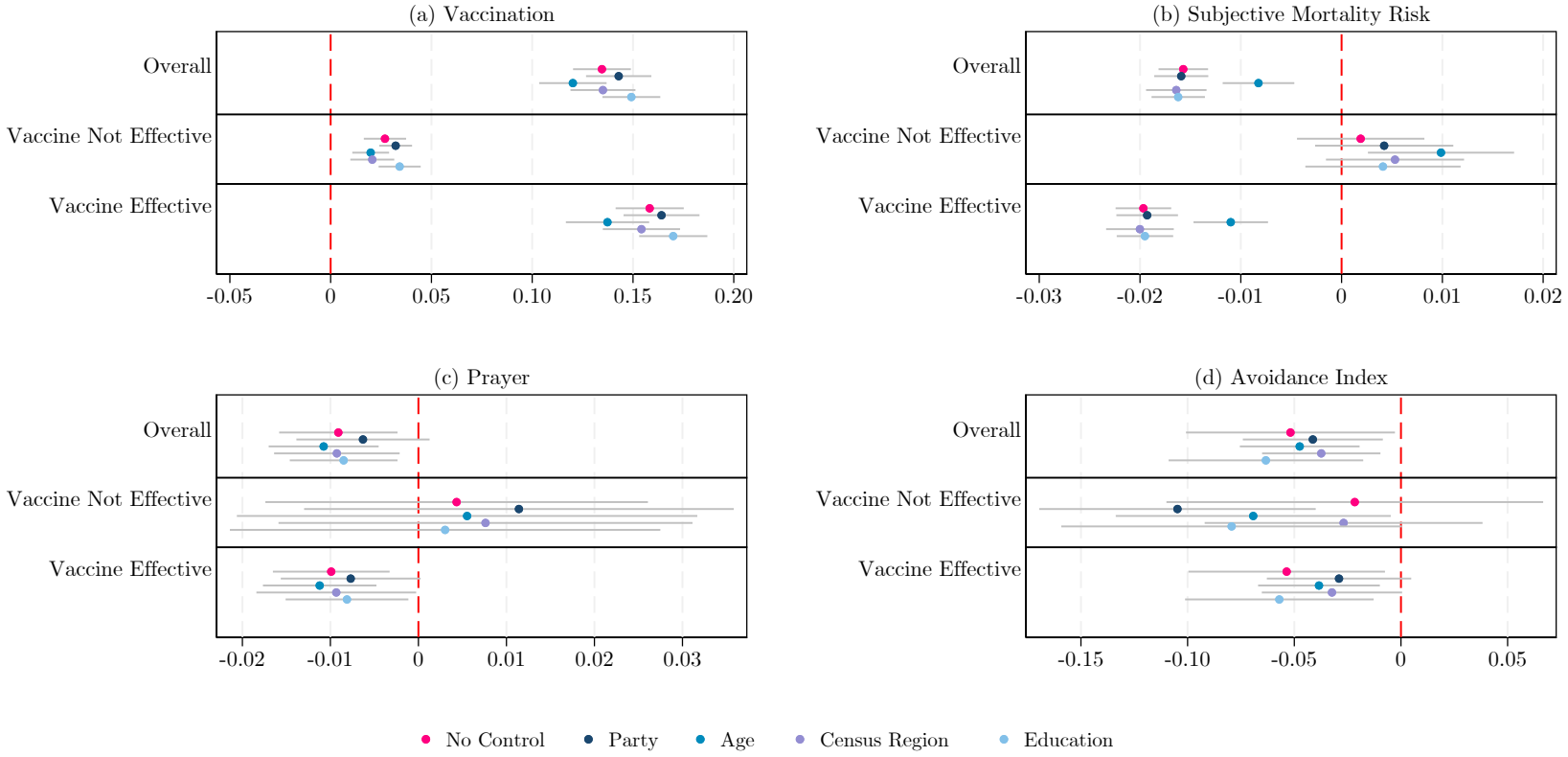


Figure 5: Estimates with Different Controls by Perceived Vaccine Effectiveness

Note: The figure presents point estimates for key outcome variables by perceived vaccine effectiveness. Within each outcome-effectiveness combination, it includes coefficients from five regressions: one with no controls and four with an interaction between the time indicator and one of the following variables: political party, age, census region, or education. In each graph, the grid line indicating a coefficient of zero is marked with a red dashed line. Error bars represent 95 percent confidence intervals based on state-clustered standard errors.

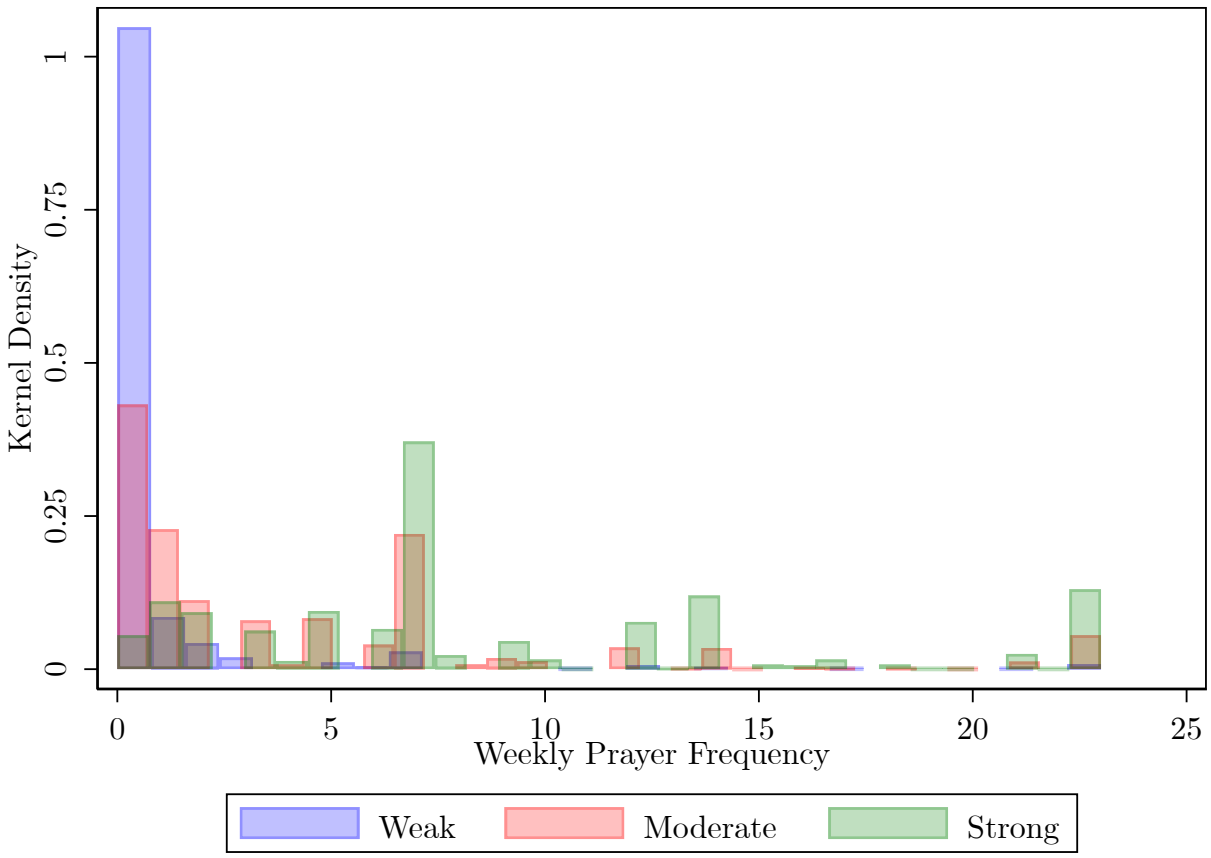


Figure 6: Distribution of Prayer Frequency By Religiosity

Note: The figure shows the frequency distribution of prayer within a week by religiosity based on survey data from December 2024 to February 2025.

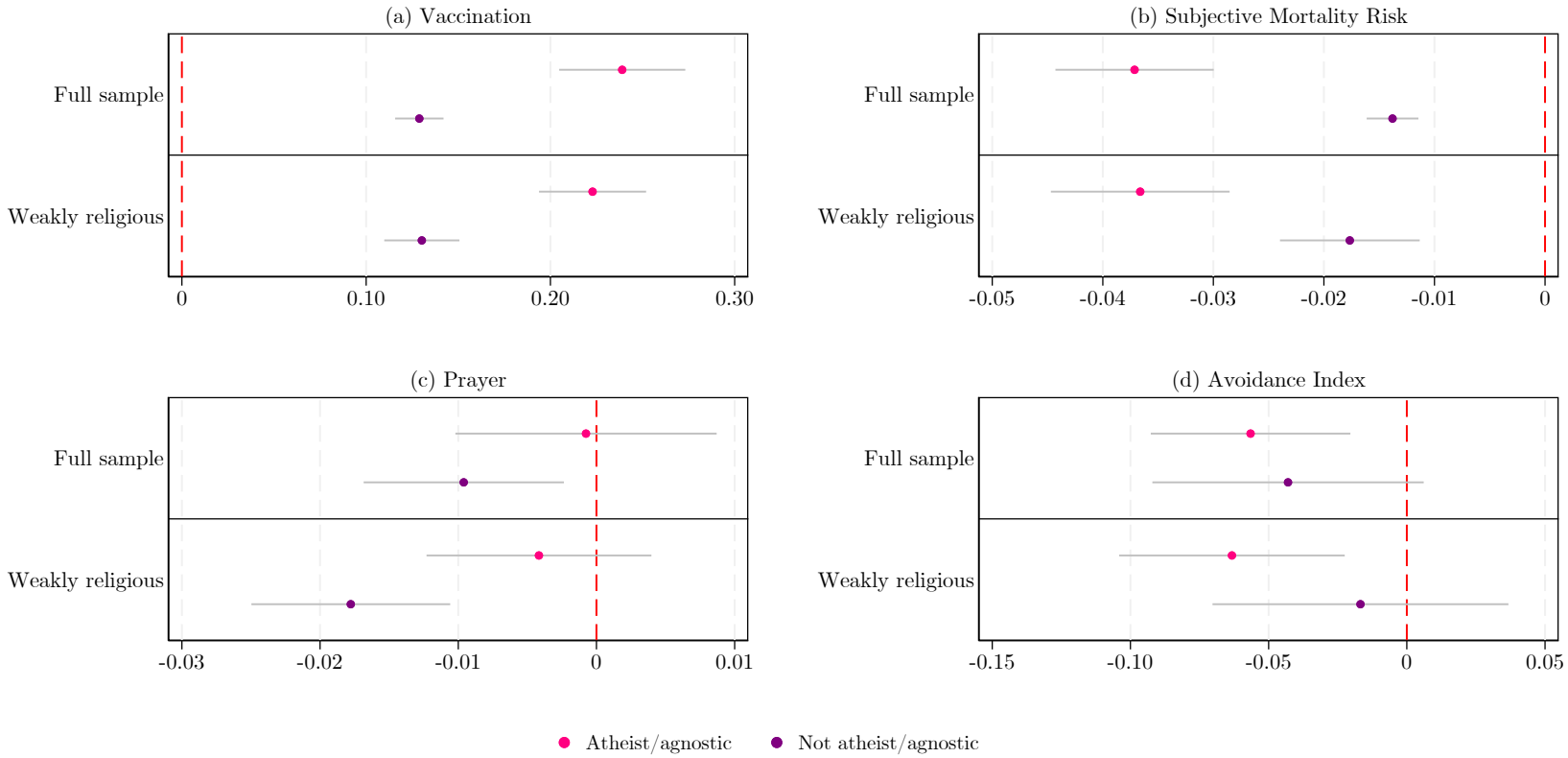


Figure 7: Estimates for Atheists and Agnostics Versus Other Respondents

Note: The figure shows the impact of vaccine eligibility by whether or not the respondent identifies as atheist or agnostic. We report estimates for the full sample (the upper band of each panel) and among weakly religious respondents (the lower band). Estimates follow the specification of Table 4, with time fixed effects interacted with perceived vaccine effectiveness and with the atheist/agnostic indicator. Error bars show 95 percent confidence intervals. Standard errors are clustered at the state of eligibility level. Estimates are weighted to be nationally representative, as described in the text.

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A Online Appendix

A.1 Additional Data Sources

To further investigate attitudes toward prayer, the survey incorporated additional variables related to religion and prayer in February and March of 2022. These questions measure whether the respondent perceives that prayer makes other COVID-19 prevention strategies, including wearing a mask and receiving a vaccination, more or less effective. An additional cross-sectional survey from December 2024 to February 2025 measures the importance of prayer for a broader array of health domains and collects more detail on the frequency and nature of prayer in general. An important caveat for these data is that they were collected four years after the vaccine rollout. Therefore it is possible that experiences with the rollout could have influenced the way that people responded to these questions.

A.2 Additional Evidence Regarding Input Substitutability

Figure A.1 presents data from the Understanding America Study in a survey collected between December 17, 2024 and February 9, 2025 (Survey UAS 673). Respondents were asked to estimate COVID-19 infection risk under hypothetical scenarios. We elicited how many out of 100 people similar to themselves would contract COVID-19 within a year when engaging in different combinations of protective behaviors, including prayer, vaccination, and non-pharmaceutical interventions such as mask use and avoiding public places. Responses were recorded on a 0–100 scale, and the order of scenarios was randomized. If two behaviors are complements, then practicing them in combination should have an effect that is larger than the sum of the effects of practicing them separately. Taken alone, vaccination reduces risk by 17 p.p., prayer reduces risk by 5 p.p., and avoidance reduces risk by 22 p.p. The combination of vaccination and prayer has a similar effect to vaccination alone. Combining vaccination and avoidance reduces risk by an additional 5 p.p. compared to vaccination alone, which is much less than the sum of the effects of the two activities. These patterns further reinforce that people do not perceive these health inputs to be complementary.

A.3 Variable Definitions

- The variable "avoidance index" incorporates social distancing and mask wearing behaviors over the past week in various contexts. The variables that are available are different in the three distinct time periods below. Within each period, we compute the first principal component of the available variables. We then standardize the combined index to have a mean of zero and a standard deviation of one across all waves.
  - **March–June 2020:** The index includes basic avoidance behaviors: (1) avoiding high-risk individuals, (2) avoiding public spaces, (3) avoiding restaurants, (4) wearing a face mask, and (5) washing hands. These questions were asked in binary form: "Yes" or "No".
  - **July 2020–June 2021:** PCA expands to include detailed avoidance behaviors, such as avoiding the following activities in the past seven days: (1) gatherings of

10+ people, (2) crowded places such as bars and clubs, (3) grocery stores or pharmacies, (4) close contact with non-household members, (5) visiting others' homes, (6) outdoor activities (e.g., walking, hiking, or exercise), (7) air travel, (8) public transportation, and (9) having visitors at home. These expanded questions had three response options: "Yes," "No," or "Unsure." The index also includes face mask usage during these activities, measured on a 6-point Likert scale: "Always," "Most of the time," "Sometimes," "Rarely," "Never," or "Unsure."

  - **September 2021–May 2023:** PCA employs only the expanded set of detailed avoidance variables and face mask usage introduced in the previous period (July 2020–June 2021), excluding the basic behavioral measures used in the initial period.

- Following the 2014 Pew Research Center Religious Landscape Study (Pew Research Center 2018), we constructed a measure of "Religiosity" as the sum of four components. Each component is categorized into three levels: high, medium, and low.

  - **Importance of Religion in Life**: This component is derived from UAS 15 (June 2015–January 2016) and UAS 37 (February 2016–October 2022). The question asked was: "How important is religion in your life?" The original question had six response options: "The most important thing," "Very important," "Somewhat important," "Not too important," "Not at all important," and "Don't know".
  - **Church Attendance**: Similar to the first component, this component is derived from UAS 15 (June 2015–January 2016) and UAS 37 (February 2016–October 2022). The question asked was: "How often do you attend religious services besides weddings and funerals?" The original question had seven response options: "Daily," "Multiple times a week," "Once a week," "One to two times a month," "Less than once a month," "Never," and "Don't know".
  - **Belief in God**: This component is derived from UAS 351 (February 2022–March 2022). The question asked was: "Which of the following comes closest to your beliefs?" The original question had four response options: "I believe in God," "I believe in some other higher power or spiritual force," "I do not believe in any higher power or spiritual force," and "I'm not sure".
  - **Prayer Frequency**: This component is constructed using responses from waves 1-21 of the UAS Coronavirus Tracking Survey. The relevant question asked: "Which of the following have you done in the last seven days to keep yourself safe from coronavirus?" "Prayer" is one of the 16 response options, and the measure is defined as the average frequency of reporting prayer across waves.

- For 27 percent of respondents, the importance or attendance component is not available from before the vaccine rollout. Table A.3 shows estimates that exclude these respondents.

A.4 Additional Estimates

Figures A.2 and A.3 distinguish categories by perceived vaccine effectiveness and religiosity, respectively. The Borusyak et al. (2024) approach does not allow us to estimate separate pre-trends by heterogeneity category, so these figures are based on subsamples.

The gradual pre-eligibility decline in perceived mortality risk in Figure 4 appears among people who perceive the vaccine as effective (Figure A.2) but not among those who do not. By religiosity (Figure A.3), it appears among moderately and strongly religious respondents but not the weakly religious.

The dip in prayer in the final pre-eligibility period comes mainly from the moderately religious (Figure A.3). Prayer among the weakly religious is flat before eligibility, so pre-trends are unlikely to explain the prayer response in this group.

Table A.2 presents estimates using alternative treatment-timing shifts, showing that the main results in Table 3 remain robust when allowing vaccination to occur up to 14 days before or after eligibility. Vaccine eligibility statistically significantly increases vaccination by 10 p.p. when the eligibility date is advanced by two weeks, and by 15 p.p. when the eligibility date is delayed by two weeks, compared to 13 p.p. when there is no shift in treatment timing. Although the estimates differ slightly when the treatment timing is changed, these differences are intuitive and small in magnitude. We therefore conclude that even if some individuals were vaccinated several days before or after their eligibility date, our estimation results remain robust. Similar patterns are observed for the other three outcomes.

Table A.3 repeats the analysis with a religiosity index that uses only information from before the vaccine rollout. The index has three components: the importance of religion, attendance at religious services, and prayer. It omits belief in God because that question was asked in 2022. We can construct this index for 73 percent of the respondents in Table 3, since the others did not answer the importance or attendance questions before the rollout. The estimates in Panel A are close to the main estimates. In Panel B, prayer falls by 1.6 p.p. for weakly religious people, by 1.1 p.p. for moderately religious people, and by 0.1 p.p. for strongly religious people ($p = 0.005$ for this difference). The differences by religiosity in vaccination and perceived mortality risk are not statistically significant ($p = 0.08$ and $p = 0.12$). For avoidance, we reject equality across the three groups ($p = 0.01$), although the estimates are not ordered by religiosity and the weakly and strongly religious groups do not differ significantly ($p = 0.33$).

Table A.4 checks whether our results depend on the choice of estimator. The Borusyak et al. (2024) estimator in our main results imputes each counterfactual from every untreated observation, which makes it precise. The Callaway and Sant'Anna (2021) estimator compares each cohort only with respondents who are not yet eligible in the same period. It relies on weaker parallel trends assumptions but is less precise.

The Callaway and Sant'Anna (2021) estimator needs one observation per person per period and a common treatment date within each cohort, so we change the data in two ways. First, we replace the daily data structure of our main estimates with a wave-level structure. We use the survey wave rather than the survey date as the time unit, keep each respondent's earliest response in a wave, and group respondents into cohorts by the wave in which they first become eligible, using three-wave bins. Second, we keep respondents in cohorts first eligible between waves 19 and 27, which retains 92 percent of the observations

in Table 3. Panel A reports the resulting estimates, with not-yet-eligible respondents as the comparison group. Vaccination and perceived mortality risk respond as in Table 3, while the estimates for prayer and avoidance are close to zero and imprecise.

Panel A differs from our main estimates in the estimator, the data structure, and the sample. Panels B and C change one of these at a time. Panel B applies the Borusyak et al. (2024) estimator to the wave-level data of Panel A, so the two panels differ only in the estimator. Panel C applies the same estimator to the same respondents but restores the daily data structure, so it differs from Table 3 only in the sample. Panel C matches Table 3 for all four outcomes, which rules out the sample restriction as the source of the gap. In Panel B, as in Panel A, prayer is not statistically significant and avoidance is significant only at the 10 percent level. The loss of significance thus comes from the wave-level data structure, not from the estimator or the sample.

Figure A.4 decomposes the Avoidance Index into 22 individual components, showing that the decrease in the Avoidance Index is primarily driven by decreases in engagement and mask-wearing behaviors when in close contact with non-household members and when attending gatherings with more than 10 people.

In the paper, we collapse perceived vaccine effectiveness into two categories. The underlying variable has four response categories, and respondents who did not answer the question form a fifth group. To assess the robustness of this approach, Figure A.6 shows estimates of the effects of vaccine eligibility for all possible vaccine effectiveness categories. The resulting patterns are broadly consistent with those obtained using the two-category classification in Table 3 and Table 5. Similarly, we collapse religiosity into three categories, however Figure A.7 shows the full distribution of the index. To assess the sensitivity to this categorization, Figure A.8 presents estimates by nine categories of religiosity, yielding patterns comparable to those based on the three-category classification in Table 4 and Table 5.

A.5 Impacts on COVID-19 infection

Table A.1 reproduces our main estimates for COVID-19 infection. Respondents may be coded as positive for COVID-19 through three possible channels: a positive COVID-19 test, a doctor's diagnosis, or a subjective belief that they contracted COVID-19. This flexibility is important because the availability of COVID-19 tests varies over the course of the pandemic. Patterns in the table are not consistent with estimates for perceived mortality risk in the paper. In Panel A, we find no overall effect of vaccine eligibility on COVID-19 infections. This pattern could reflect the fact that vaccines did more to reduce the severity of COVID-19 infections rather than to prevent them. In Panel B, vaccine eligibility reduces COVID-19 infections significantly for those who perceive that the vaccine is not effective, while it appears to have no effect on infections among those who perceive the vaccine to be effective. The difference between the change of these two groups of people is statistically significant ($p < 0.01$). Finally, in Panel C, we find no significant effects for any religiosity subgroups. Results are not consistent with our findings in Table 4, which shows that all groups reduce perceived COVID-19 mortality risk.

A.6 Determination of Vaccine Eligibility

We calculate vaccine eligibility by combining individual characteristics with state-specific eligibility criteria. As described in the text, the federal government purchased vaccine doses from drug manufacturers and then relied on state health departments to distribute vaccines. Following CDC guidelines, these agencies prioritized providing vaccines to first responders and vulnerable individuals, including the elderly and those with compromised health situations. States set rollout dates based on their interpretation of these guidelines, access to vaccine doses, and local infrastructure for vaccine delivery. Multistate, a government relations firm, maintained a list of state-specific eligibility dates and criteria. We combine these data with information on individual UAS respondents to determine each person's eligibility. These characteristics include age, chronic health conditions, residence in a nursing home, occupation, and state of residence. Where possible, we rely on responses that occurred prior to the vaccine rollout. However, occupation data were collected during February-April 2021, during the vaccine rollout. Occupation data are based on six-digit ONET codes, which is the highest level of granularity. We applied for access to these confidential data and utilized a secure data enclave to construct the eligibility variable.

One concern with reliance on occupation data that were collected during the vaccine rollout is that some people might sort into occupations that give them earlier eligibility. Such sorting seems unlikely given the training and experience needed to enter many of the relevant occupations (e.g., first responders, health care workers and educators). As an additional test of possible occupational sorting, we examine whether vaccine eligibility is associated with any unusual employment patterns early in the vaccine rollout. We do not find evidence of sorting. From December 11, 2020 to March 8, 2021 (the median eligibility date), 98 percent of employed early-eligible people remained in the same job while 97 percent of employed late-eligible people did so. This suggests that it is unlikely people chose occupations to hasten vaccine eligibility.

Table A.1: The Impact of Vaccine Eligibility on COVID-19 Infection

	COVID-19
	(1)
A: Overall Impact	
Overall	-0.0020
	(0.0020)
B: Impact by Perceived Vaccine Effectiveness	
Vaccine not effective	-0.019***
	(0.0047)
Vaccine effective	0.0023
	(0.0020)
Equality of coefficients (p-value)	0.00
C: Impact by Religiosity	
Weakly religious	-0.0027
	(0.0036)
Moderately religious	-0.0011
	(0.0014)
Strongly religious	-0.0037
	(0.0032)
Equality of coefficients (p-value)	0.62
Weak vs. strong (p-value)	0.79

Note: Regressions are based on the specification: $Y_{it} = \mu_i + \lambda_{te} + \delta E_{it} + \epsilon_{it}$ in which i indexes the individual, t indexes the survey date, e is perceived vaccine effectiveness, and E_{it} is the individual's vaccine eligibility. Estimates follow the difference-in-difference imputation approach of Borusyak et al. (2024). Estimates are weighted to be nationally representative in terms of sex, race, age, marital status, and education, as explained in the text. Standard errors are clustered at the state of eligibility level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A.2: The Impact of Vaccine Eligibility with a Shifted Treatment Timing

	Vaccination	Mortality Risk	Prays Weekly	Avoidance Index
	(1)	(2)	(3)	(4)
A: Vaccine Eligibility Date Advanced by 14 Days				
Overall	0.10***	-0.012***	-0.0067*	-0.058***
	(0.0065)	(0.0012)	(0.0036)	(0.019)
Observations	180,278	180,278	180,278	161,498
B: Vaccine Eligibility Date Delayed by 14 Days				
Overall	0.15***	-0.017***	-0.0089**	-0.047**
	(0.0083)	(0.0016)	(0.0037)	(0.022)
Observations	183,569	183,569	183,569	164,295
C: Drop Observations from 14 to 1 Days Before Eligibility				
Overall	0.13***	-0.015***	-0.0072*	-0.065***
	(0.0081)	(0.0012)	(0.0043)	(0.018)
Observations	175,807	175,807	175,807	157,531

Note: Regressions are based on the specification: $Y_{it} = \mu_i + \lambda_{te} + \delta E_{it} + \epsilon_{it}$ in which i indexes the individual, t indexes the survey date, e is perceived vaccine effectiveness, and E_{it} is the individual's vaccine eligibility. In the regressions, the treatment date E_{it} is hypothetically shifted 14 days earlier or later for each treated unit. Panel C drops observations from the 14 days before eligibility. Standard errors are clustered at the state of eligibility level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A.3: The Impact of Vaccine Eligibility Using the Pre-Rollout Religiosity Index

	Vaccination	Mortality Risk	Prays Weekly	Avoidance Index
	(1)	(2)	(3)	(4)
A: Overall Impact				
Overall	0.13***	-0.013***	-0.0094**	-0.044*
	(0.0079)	(0.0014)	(0.0039)	(0.024)
Pre-rollout mean	0.00	0.19	0.60	-0.04
Observations	138,212	138,212	138,212	124,025
B: Impact by Religiosity				
Weakly religious	0.15***	-0.019***	-0.016***	-0.062**
	(0.0099)	(0.0033)	(0.0042)	(0.026)
Moderately religious	0.12***	-0.0091***	-0.011*	-0.0052
	(0.014)	(0.0032)	(0.0064)	(0.021)
Strongly religious	0.14***	-0.013***	-0.0011	-0.043
	(0.0074)	(0.0020)	(0.0040)	(0.028)
Pre-rollout mean				
Weakly religious	0.00	0.16	0.10	-0.03
Moderately religious	0.00	0.22	0.75	0.01
Strongly religious	0.00	0.18	0.95	-0.12
Equality of coefficients (p-value)	0.08	0.12	0.00	0.01
Weak vs. strong (p-value)	0.27	0.12	0.01	0.33
Observations	138,175	138,175	138,175	123,992

Note: Regressions are based on the specification: $Y_{it} = \mu_i + \lambda_{te} + \delta E_{it} + \epsilon_{it}$ in which i indexes the individual, t indexes the survey date, e is perceived vaccine effectiveness, and E_{it} is the individual's vaccine eligibility. The religiosity index is the sum of three components: the importance of religion, attendance at religious services, and prayer. Each component is measured using only survey waves that were completed before December 11, 2020, the first date on which any respondent became eligible for a vaccine. Belief in God is omitted from the index because it was measured in 2022. The sample retains 75 percent of the observations and 73 percent of the respondents in Table 3. In Panel B, time indicators are additionally interacted with religiosity category. Estimates follow the difference-in-difference imputation approach of Borusyak et al. (2024). Estimates are weighted to be nationally representative in terms of sex, race, age, marital status, and education, as explained in the text. Standard errors are clustered at the state of eligibility level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A.4: Callaway and Sant'Anna (2021) Robustness Checks

	Vaccination	Mortality Risk	Prays Weekly	Avoidance Index
	(1)	(2)	(3)	(4)
A: CS Estimator				
Overall	0.12***	-0.013***	0.0023	-0.022
	(0.019)	(0.0036)	(0.0051)	(0.023)
Observations	134,505	134,505	134,505	124,043
B: BJS Estimator, CS Data Structure				
Overall	0.12***	-0.0072**	-0.0059	-0.038*
	(0.012)	(0.0031)	(0.0056)	(0.020)
Observations	138,645	138,645	138,645	128,970
C: BJS Estimator, Original Data Structure				
Overall	0.13***	-0.016***	-0.0091**	-0.056**
	(0.0073)	(0.0012)	(0.0036)	(0.024)
Equality with Table 3 (p-value)	0.68	0.96	0.99	0.90
Observations	168,170	168,170	168,170	149,793

Note: All panels restrict the sample to respondents first eligible in waves 19 to 27, grouped into three cohorts (waves 19 to 21, 22 to 24, and 25 to 27). Wave 19 is the first wave of the vaccine rollout. Respondents first eligible in waves 28 to 34 are excluded because no not-yet-eligible respondents remain to compare them with. Panel A reports the aggregate average treatment effect on the treated (ATT) from the Callaway and Sant'Anna (2021) estimator, with not-yet-eligible cohorts as the comparison group and perceived vaccine effectiveness as a covariate. Panels A and B use a wave-level data structure: one observation per person per survey wave (the earliest response), the survey wave as the time unit, and eligibility timing defined at the cohort level. Panel B applies the Borusyak et al. (2024) imputation estimator to the same data as Panel A. Panel C applies the Borusyak et al. (2024) estimator to the same respondents using the daily data structure of Table 3: all responses, the survey date as the time unit, and individual eligibility dates. Panels A and B have different numbers of observations (134,505 and 138,645) because the Callaway and Sant'Anna (2021) estimator drops cohort-wave cells without not-yet-eligible comparison respondents, while the Borusyak et al. (2024) estimator keeps every observation that contributes to the imputation. "Equality with Table 3" reports p-values from tests that each Panel C estimate equals the corresponding estimate in Table 3. Standard errors, clustered by state of residence at the time of eligibility, are in parentheses. Estimates are weighted to be nationally representative by sex, race, age, marital status, and education. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A.5: Pre-Rollout Religious Denominations by Religiosity and Vaccine Eligibility

		Religiosity			Vaccine Eligibility		P-values	
	All	Weak	Moderate	Strong	Early	Late	Relig.	Elig.
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Evangelical Protestant	0.17	0.02	0.13	0.31	0.18	0.15	0.00	0.02
Non-Evangelical Protestant	0.18	0.06	0.19	0.24	0.20	0.15	0.00	0.00
Catholic	0.18	0.08	0.21	0.21	0.20	0.16	0.00	0.04
Other religion	0.18	0.12	0.20	0.19	0.16	0.20	0.00	0.00
Atheist or agnostic	0.08	0.26	0.05	0.00	0.07	0.09	0.00	0.01
No particular religion	0.21	0.46	0.21	0.05	0.18	0.25	0.00	0.00
Number of Observations	139,400	33,351	60,911	45,138	66,107	73,243	–	–

Note: The table reports sample means for the period from March 11-December 10, 2020. All statistics use post-stratification weights to ensure that the sample is representative of US adults in terms of gender, race, age, marital status, and education, as explained in the text. Vaccine eligibility categories are split at the median of March 8, 2021. Religious identity is taken from each respondent's earliest available response: UAS 308 (August-September 2020) or UAS 318 (November-December 2020) for 89 percent of respondents, otherwise UAS 393 (June-July 2021) or wave 32 of the COVID-19 survey (July-September 2022). Shares are computed among respondents with a recorded identity (97 percent of observations) and sum to one within each column. Atheists and agnostics exclude respondents reporting "no particular religion," who are shown separately. P-values test equality across religiosity categories (column 7) and across eligibility groups (column 8), using standard errors clustered at the state of eligibility level.

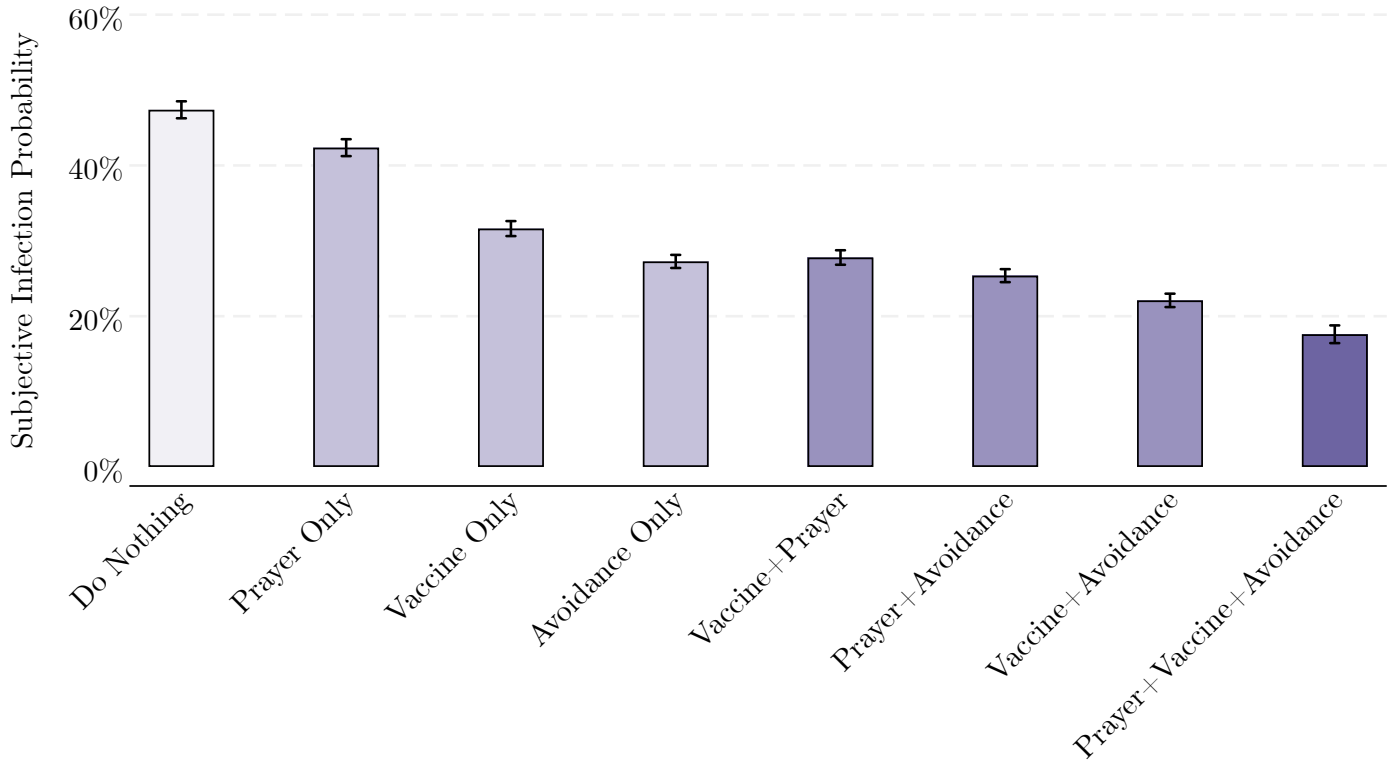


Figure A.1: Perceptions About the Complementarity Between COVID-19 Prevention Behaviors

Note: The figure presents point estimates for subjective infection probability when using different preventive behaviors.

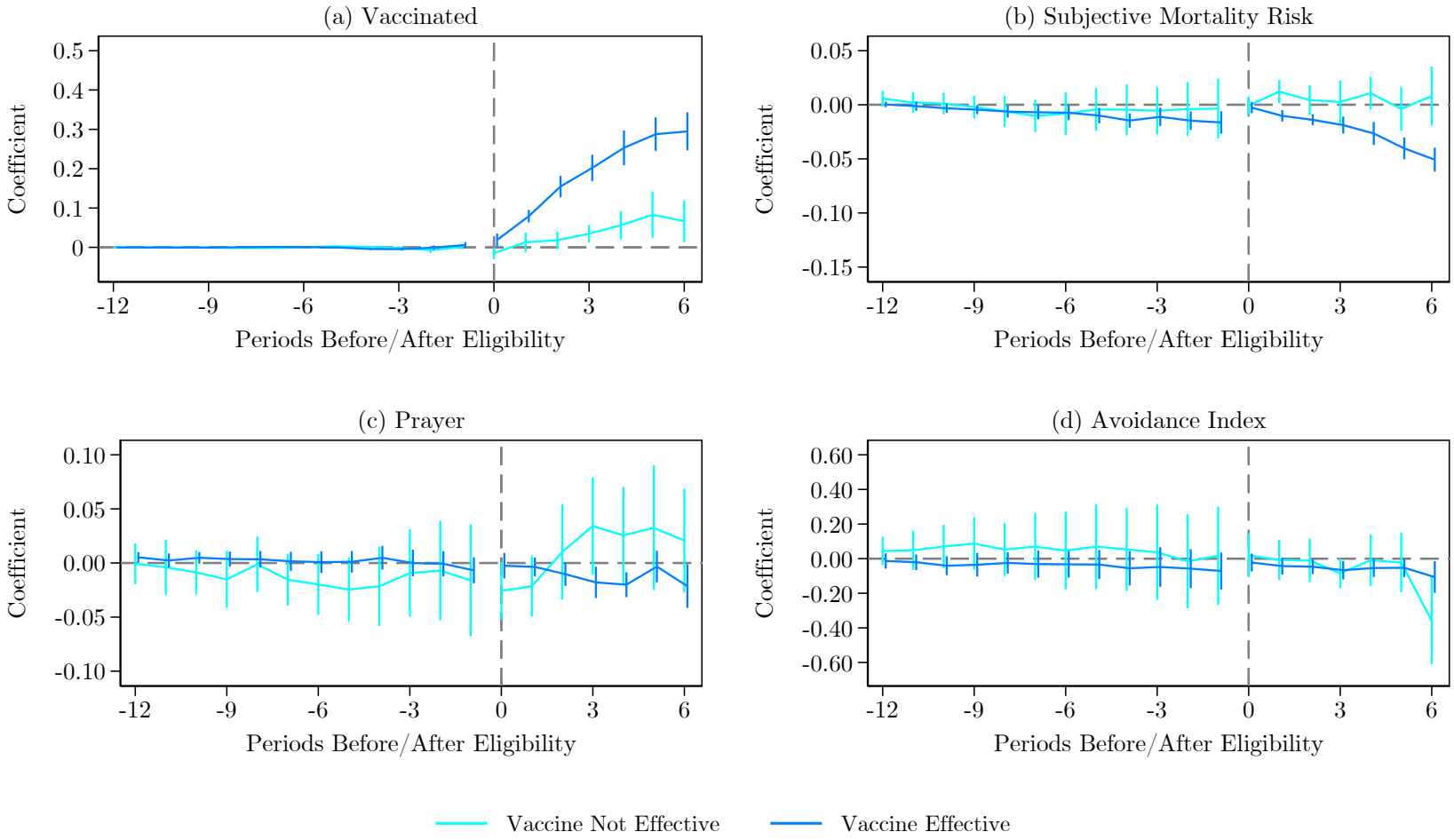


Figure A.2: Trends by Perceived Vaccine Effectiveness Using Subsamples

Note: The figure estimates biweekly pre- and post-eligibility trends by perceived vaccine effectiveness for key outcome variables. Estimates are based on the DID imputation method of Borusyak et al. (2024). Variables are defined as in the text. Error bars indicate 95 percent confidence intervals based on state-clustered standard errors. Estimates are weighted to be nationally representative. Pre-eligibility coefficients are measured relative to periods more than 24 weeks before eligibility, while post-eligibility coefficients are measured relative to the imputed counterfactual, so the two series do not share a baseline. The post-eligibility estimates assume parallel trends and do not net out any pre-existing differential trend.

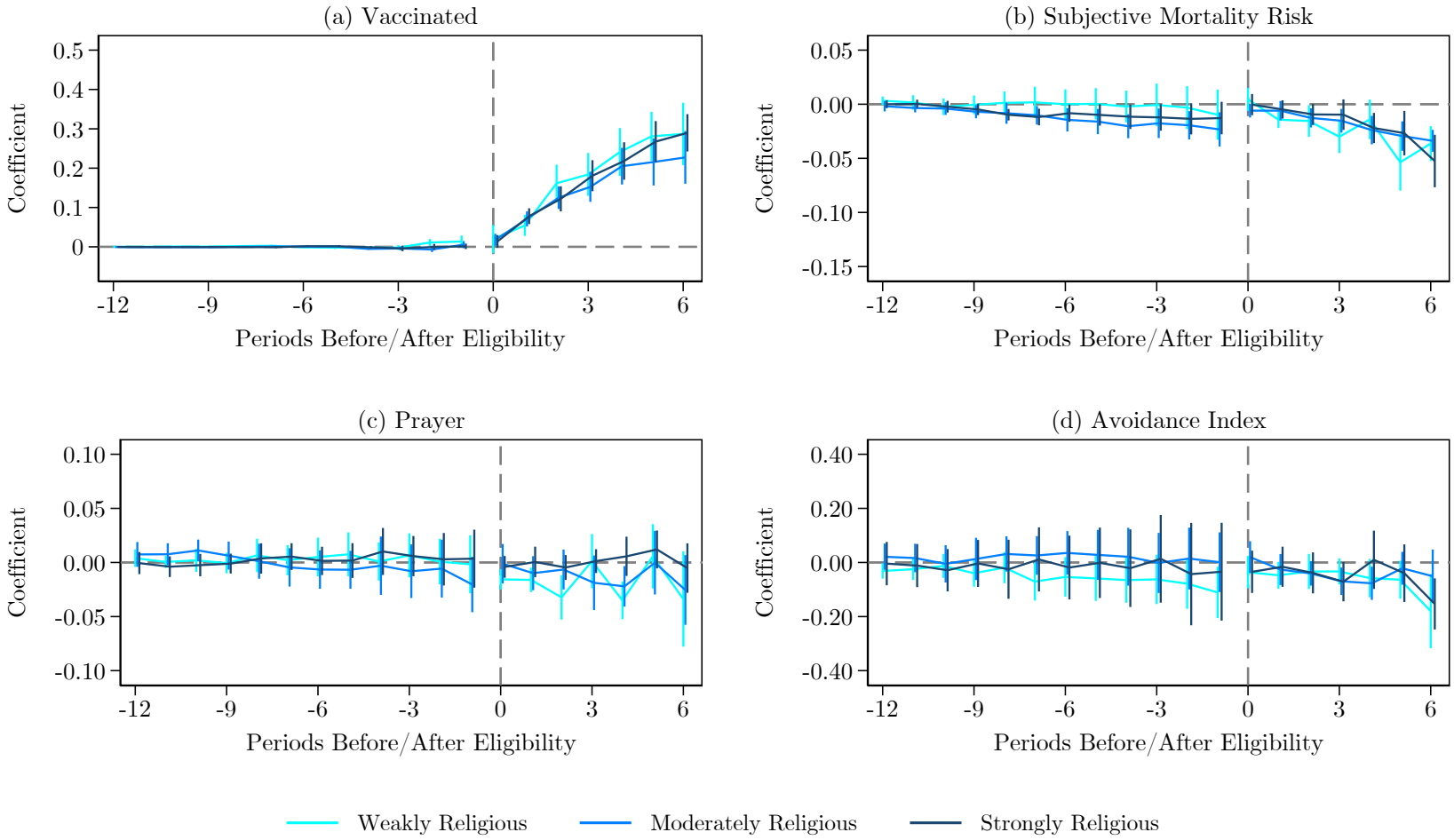


Figure A.3: Trends by Religiosity Using Subsamples

Note: The figure estimates biweekly pre- and post-eligibility trends by religiosity for key outcome variables. Estimates are based on the DID imputation method of Borusyak et al. (2024). Variables are defined as in the text. Error bars indicate 95 percent confidence intervals based on state-clustered standard errors. Estimates are weighted to be nationally representative. Pre-eligibility coefficients are measured relative to periods more than 24 weeks before eligibility, while post-eligibility coefficients are measured relative to the imputed counterfactual, so the two series do not share a baseline. The post-eligibility estimates assume parallel trends and do not net out any pre-existing differential trend.

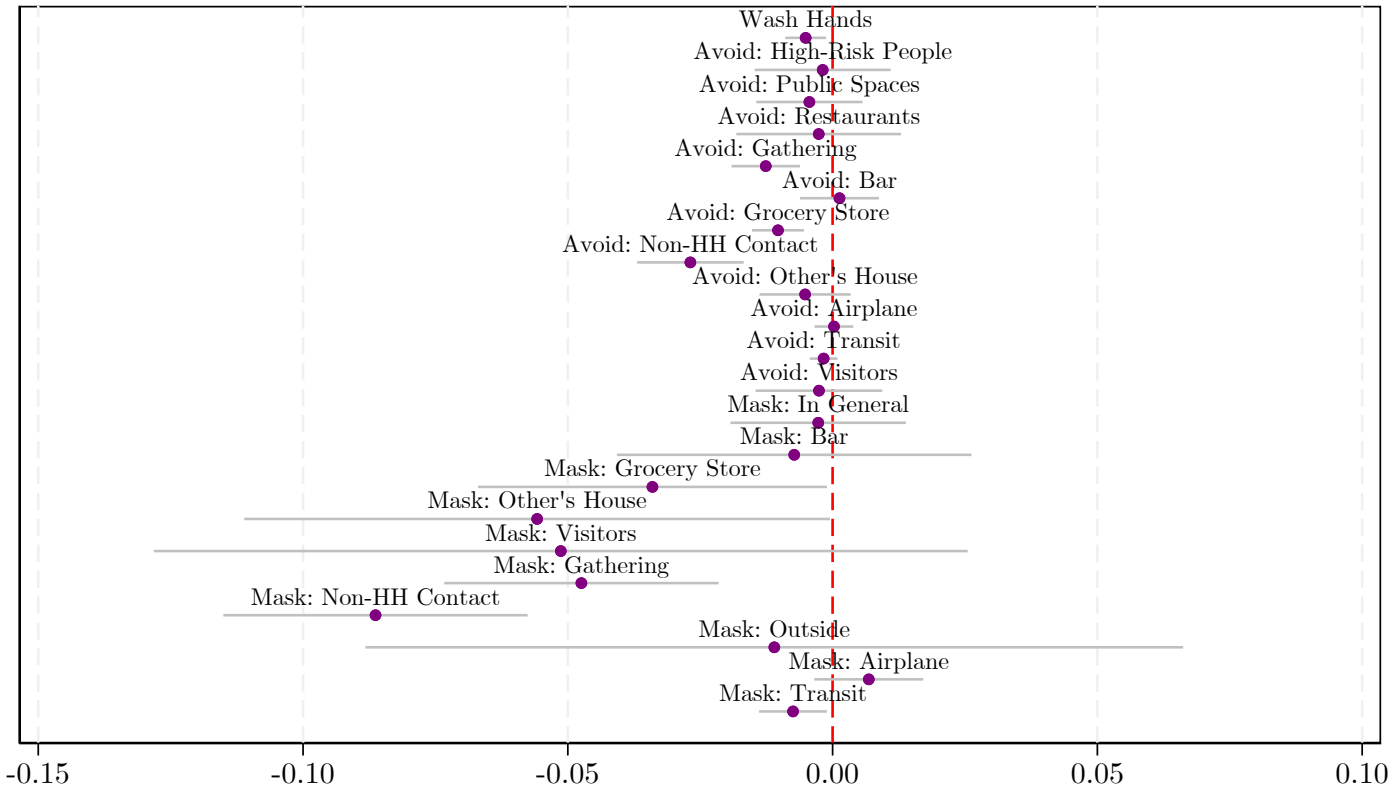


Figure A.4: Effects of Vaccine Eligibility on Components of Avoidance Index

Note: This figure shows the estimated impact of vaccine eligibility on the individual components of the Avoidance Index. The index captures a range of behaviors, including avoiding high-risk individuals, public spaces, restaurants, and social gatherings, as well as engaging in personal protective practices such as mask wearing and handwashing.

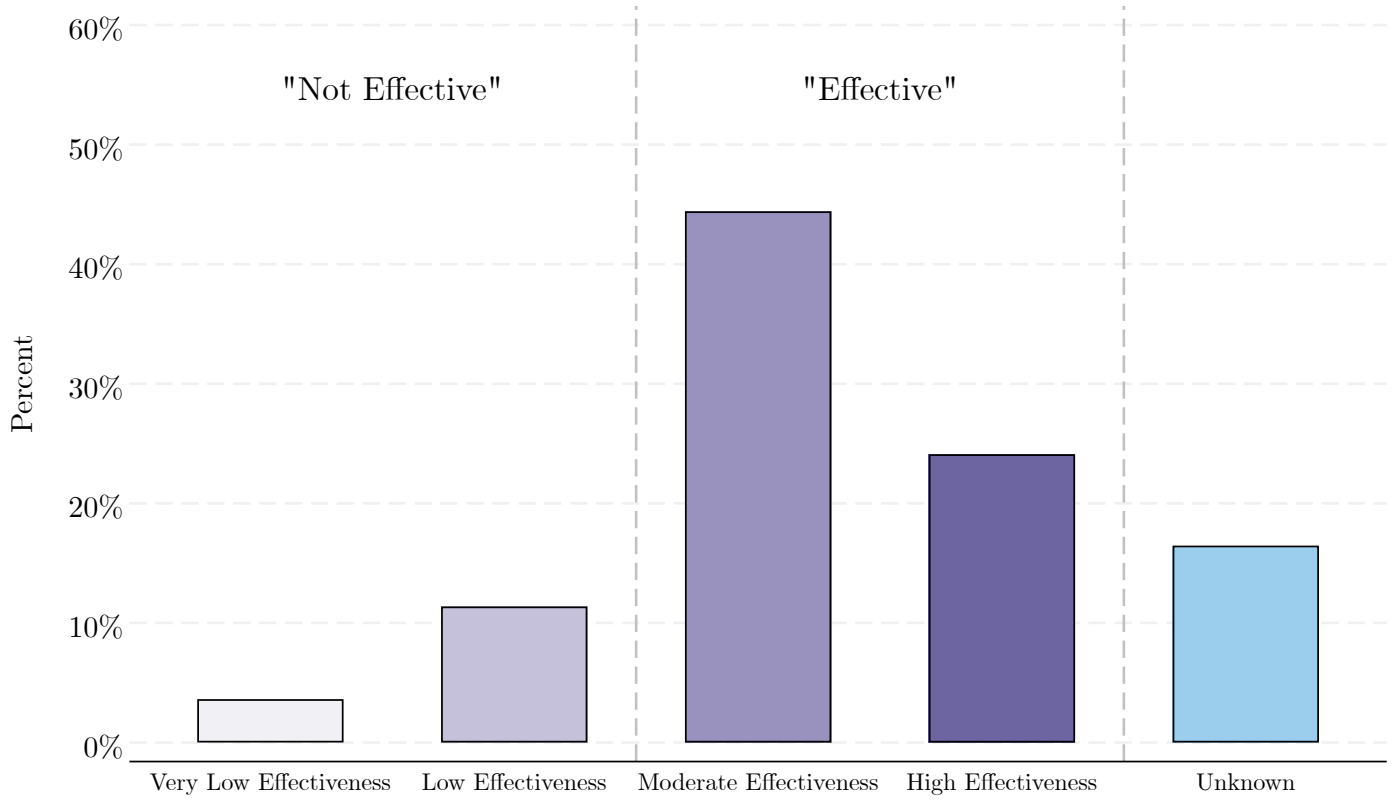


Figure A.5: Distribution of Effectiveness of Vaccine

Note: The figure presents the distribution of respondents' perceived vaccine effectiveness across the full set of five categories and illustrates how these were collapsed into two groups for the main analysis.

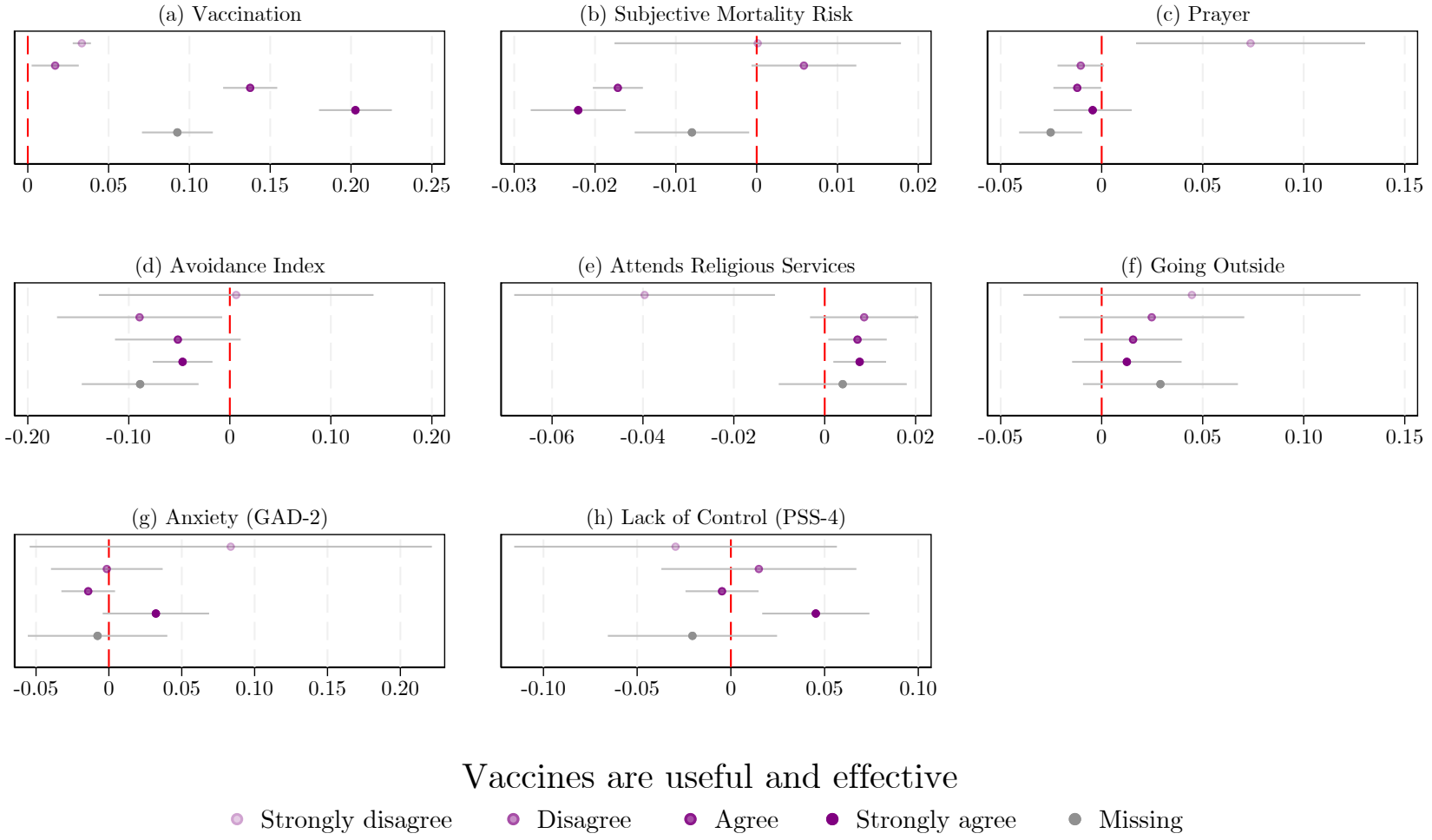


Figure A.6: Effects of Vaccine Eligibility on Outcomes by Full Categories of Effectiveness of Vaccine

Note: The figure shows the estimates for the main outcomes across the five categories of perceived vaccine effectiveness.

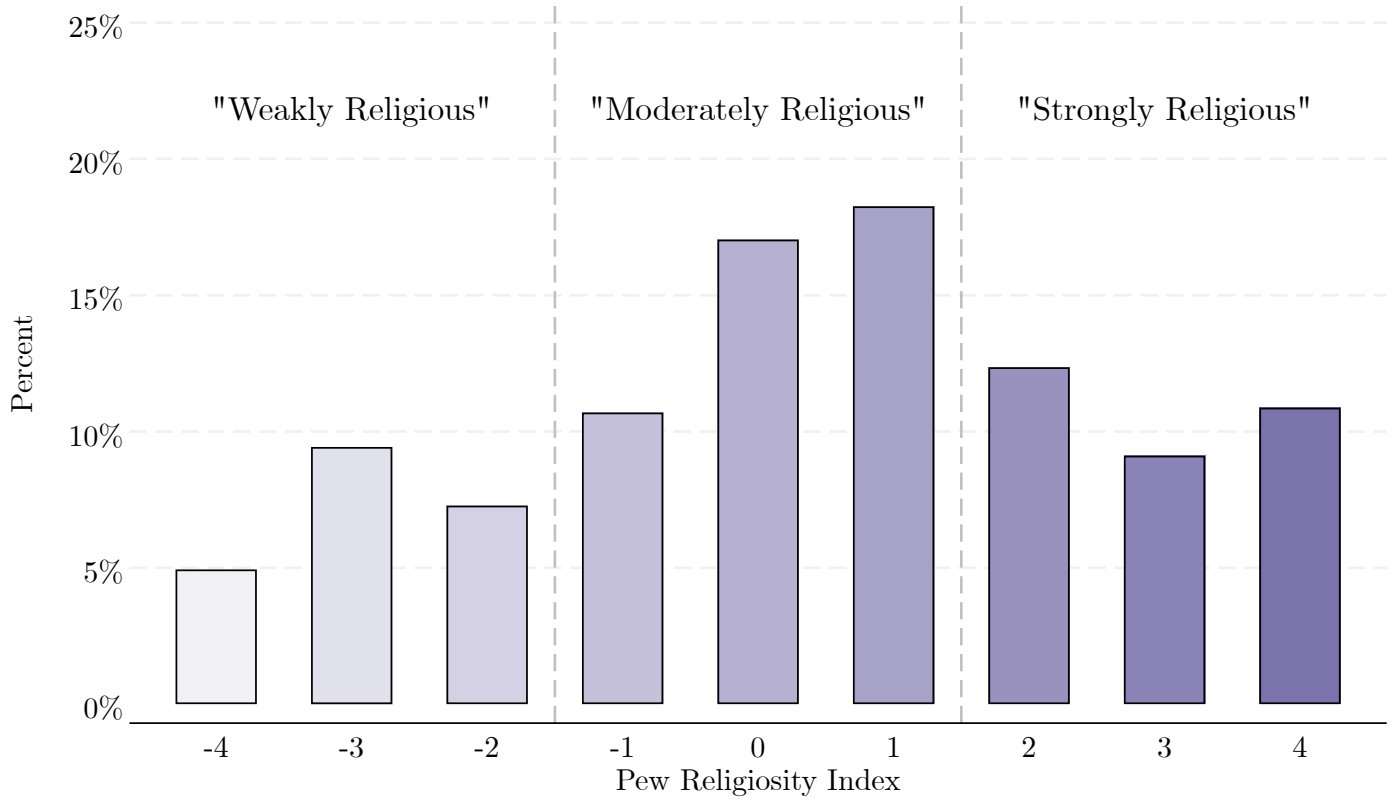


Figure A.7: Distribution of Religiosity

Note: The figure presents the distribution of respondents' religiosity across the full set of nine categories and illustrates how these were collapsed into three groups for the main analysis.

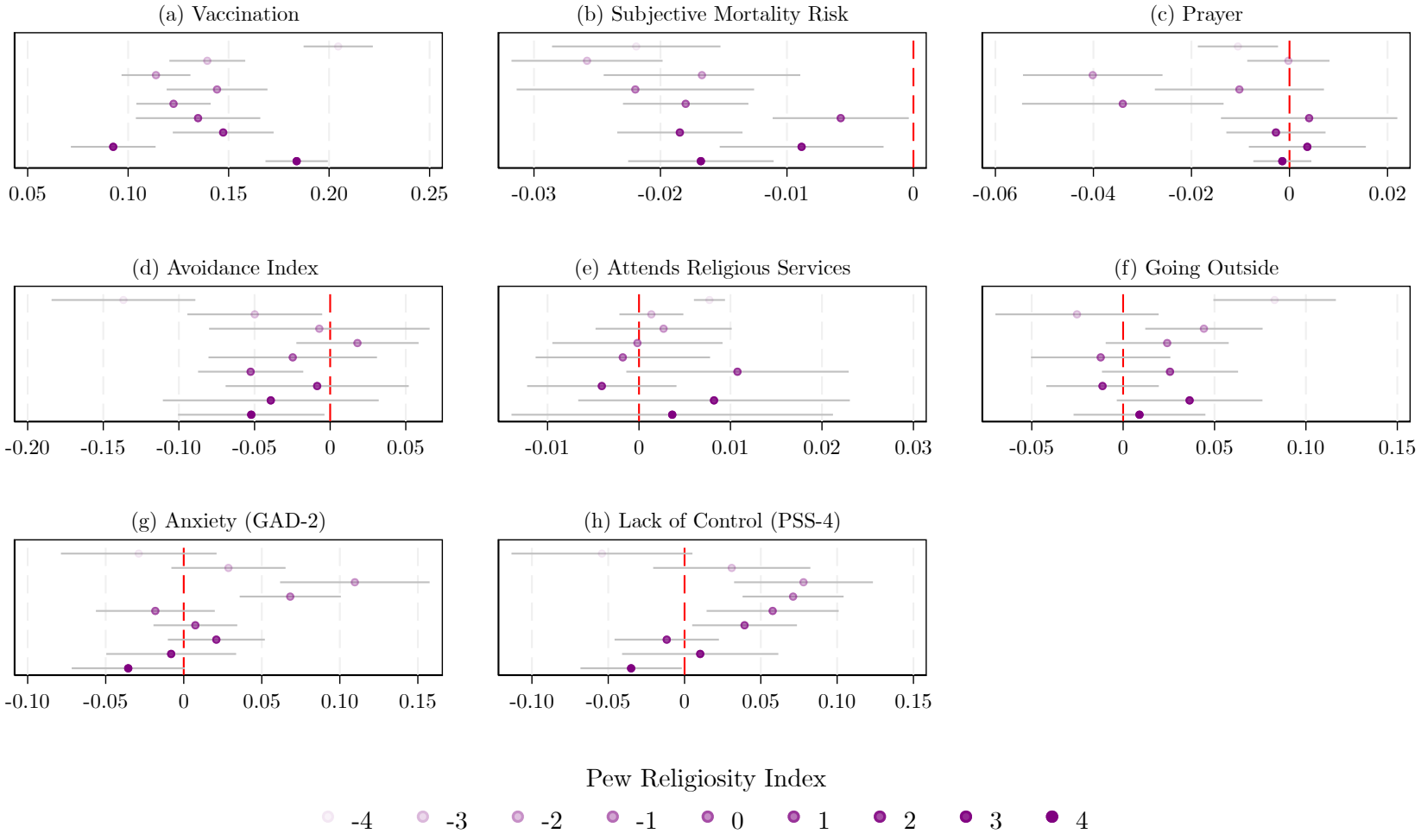


Figure A.8: Effects of Vaccine Eligibility on Outcomes by Full Categories of Religiosity

Note: The figure shows the estimates for the main outcomes across the nine categories of religiosity.

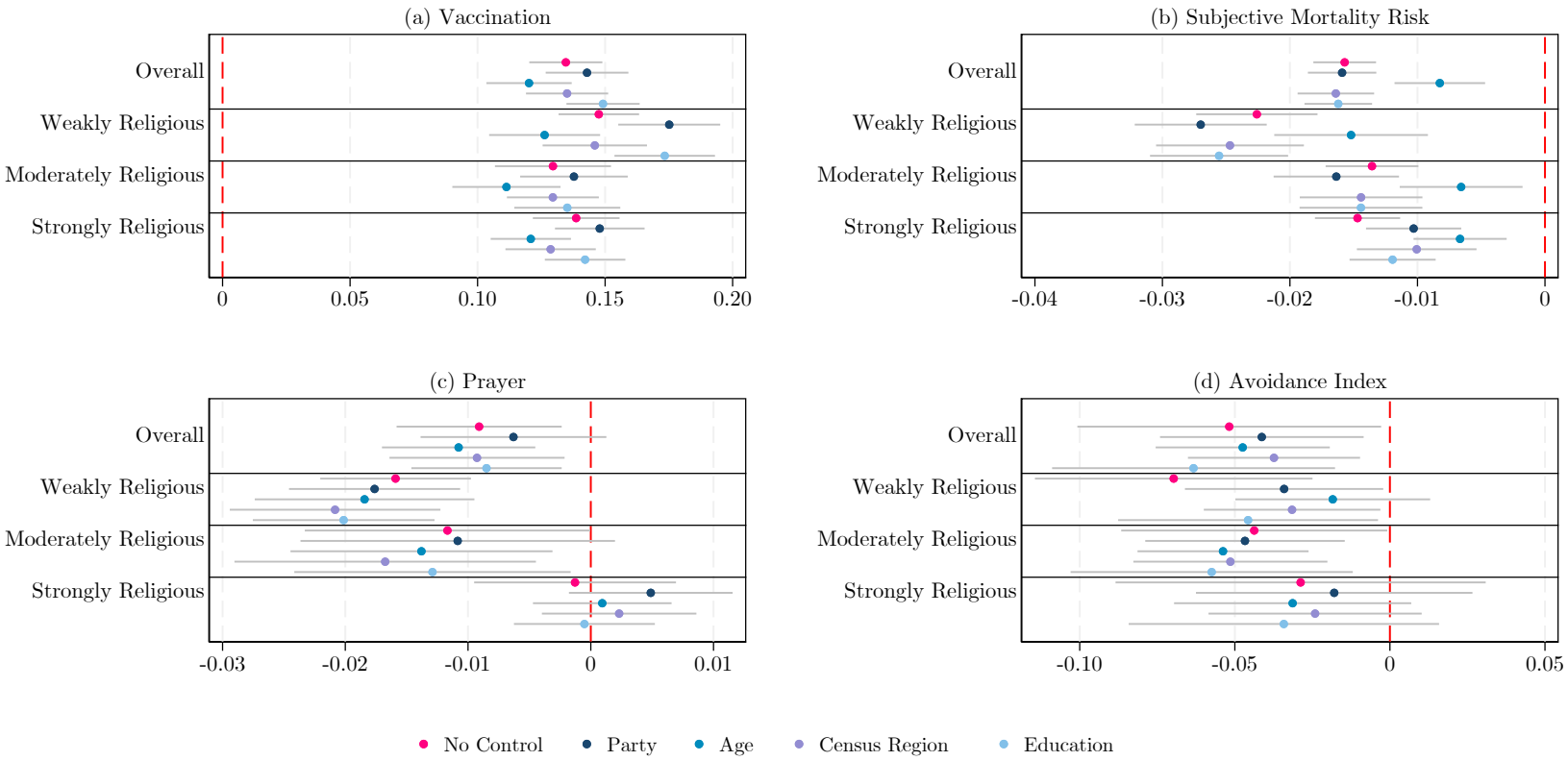


Figure A.9: Estimates with Different Controls by Religiosity

Note: The figure presents point estimates for key outcome variables by religiosity. Within each outcome-religiosity combination, it includes coefficients from five regressions: one with no controls and four with an interaction between the time indicator and one of the following variables: political party, age, census region, or education. In each graph, the grid line indicating a coefficient of zero is marked with a red dashed line. Error bars represent 95 percent confidence intervals based on state-clustered standard errors.